Distribution-Valued Subsequential Limits of Regularized Logarithms of the Critical 2D SHF

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Abstract

We prove compactness, modulo additive spatial constants, for regularized logarithms of the critical two-dimensional stochastic heat flow. Fix a time interval $s < t$, a nonzero nonnegative $g \in C_c^\infty(\mathbb{R}^2)$, and an admissible mollifier ρ. For

$$H_\varepsilon^{g,\rho}(y) = \log \iint g(x)\rho_\varepsilon(y-z)Z_{s,t}^\theta(\,\mathrm{d}x,\,\mathrm{d}z), \qquad c_\varepsilon = \frac{1}{2}\log\bigl(1+\log\varepsilon^{-2}\bigr),$$

the laws of $H_\varepsilon^{g,\rho} + c_\varepsilon$ are tight in the strong distribution space $\mathcal{D}'_\beta$. Consequently the quotient fields $[H_\varepsilon^{g,\rho}]$ are tight in $\mathcal{D}'_{0,\beta} \simeq \mathcal{D}'_\beta/\operatorname{span}\{1\}$, and every sequence $\varepsilon \downarrow 0$ has a subsequence converging in law to a Radon random distribution modulo constants.

The proof combines the local signed-chain estimates of Gu–Tsai with a complete multiscale interval-gas argument, denominator taming, Chen's Gaussian convexity, exact two-replica second moments, and a nuclear-space tightness criterion of Fouque. The result is a compactness theorem only: uniqueness of the subsequential law and mollifier independence are not asserted.

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1 Introduction

1.1 Main results

The critical two-dimensional stochastic heat flow (SHF), denoted by $Z^{\theta}_{s,t}(\,\mathrm{d}x,\,\mathrm{d}z)$, is a random measure-valued flow introduced by Caravenna, Sun, and Zygouras [2]. We use the normalization and structural properties recorded in Tsai [5] and Gu–Tsai [6]. Throughout,

$$p_t(x) = \frac{1}{2\pi t}\exp\!\left(-\frac{|x|^2}{2t}\right)$$

denotes the two-dimensional heat kernel.

An *admissible mollifier* is a function

$$\rho \in C_c^{\infty}(\mathbb{R}^2), \qquad \rho \geq 0, \qquad \int_{\mathbb{R}^2} \rho(x)\,\mathrm{d}x = 1,$$

which is strictly positive on a neighborhood of the origin. Set

$$\rho_\varepsilon(x) = \varepsilon^{-2}\rho(x/\varepsilon), \qquad \check{\rho}(x) = \rho(-x).$$

Fix $s < t$, a nonzero nonnegative $g \in C_c^{\infty}(\mathbb{R}^2)$, and a fixed admissible ρ. Define the regularzied logarithm of the SHF by

$$H^{g,\rho}_{\varepsilon}(y) = \log \iint g(x)\rho_\varepsilon(y-z)Z^{\theta}_{s,t}(\,\mathrm{d}x,\,\mathrm{d}z).$$

We also set

$$\ell_\varepsilon := 1 + \log\varepsilon^{-2}, \qquad c_\varepsilon := \frac{1}{2}\log\ell_\varepsilon, \qquad q(\varphi) := \|\varphi\|_{L^2} + \int_{\mathbb{R}^2}(1+|y|^2)|\varphi(y)|\,\mathrm{d}y. \tag{1}$$

Let $\mathcal{D} = C_c^{\infty}(\mathbb{R}^2)$ with its usual strict LF topology and

$$\mathcal{D}_0 = \left\{\varphi \in \mathcal{D} : \int_{\mathbb{R}^2}\varphi = 0\right\}.$$

Write $\mathcal{D}'_\beta$ for the strong dual of $\mathcal{D}$ and $\mathcal{D}'_{0,\beta}$ for the strong dual of $\mathcal{D}_0$; canonically,

$$\mathcal{D}'_{0,\beta} \simeq \mathcal{D}'_\beta/\operatorname{span}\{1\}.$$

The purpose of the main theorem is to construct a compact family of "height-type" logarithmic observables associated with the critical SHF. The pointwise logarithm has a diverging deterministic component, so the natural compactness statement is distribution-valued and, ultimately, modulo spatial constants. Establishing this compactness is the first step toward identifying a canonical limiting logarithmic field, which serves as a candidate solution of the 2d KPZ.

The following theorem is the central quantitative estimate. Its role is to give a uniform first-moment bound for every test function; Section 6 then upgrades this scalar control to tightness of random distributions.

Theorem 1.1 (Compactness for the original regularizations)**.** *For every fixed $s < t$, g, and ρ as above, there are $C < \infty$ and $\varepsilon_0 > 0$ such that, for every real $\varphi \in C_c^{\infty}(\mathbb{R}^2)$,*

$$\boxed{\sup_{0<\varepsilon<\varepsilon_0} \mathbb{E}\,|\langle H^{g,\rho}_{\varepsilon} + c_\varepsilon, \varphi\rangle| \leq Cq(\varphi).} \tag{2}$$

The following corollary records the topological consequence of Theorem 1.1: the regularized logarithms have subsequential limits as random distributions, after removing the deterministic constant mode.

Corollary 1.2 (Tightness of regularized logarithms)**.** *The family* $\{H_\varepsilon^{g,\rho} + c_\varepsilon\}$ *is tight in* $\mathcal{D}'_\beta$*. In particular,* $\{[H_\varepsilon^{g,\rho}]\}$ *is tight in* $\mathcal{D}'_{0,\beta} \simeq \mathcal{D}'_\beta / \operatorname{span}\{1\}$.

Remark 1.3. Neither result asserts uniqueness of the limit, comparison errors tending to zero for different mollifiers, or convergence on the original probability space.

Remark 1.4. To see that $H_\varepsilon^\rho = \log F_\varepsilon^\rho$ is well defined, we verify here that F_ε^ρ is finite and postivie. Since $\mathbb{E}[\mu(K)] < \infty$ for all compact set K, we know $\mu(\,\mathrm{d}y)$ is locally finite and thus $F_\varepsilon^\rho(y)$ is finite a.s. For strict local positivity of $F_\varepsilon^\rho(y)$, apply Nakashima's Theorem 1.4 [8] with initial profile $u_0 = g$; the conditional-GMC approach of Clark–Tsai [7] gives an independent route. For each rational ball the resulting μ-mass is almost surely positive. Taking the countable intersection over rational balls yields one probability-one event on which every rational ball has positive μ-mass, and hence every nonempty open ball does.

Next, we check $H_\varepsilon^\rho = \log F_\varepsilon^\rho$ is smooth and locally integrable. For each compact K, the kernels $\rho_\varepsilon(y - \cdot)$, $y \in K$, are supported in one compact set. Differentiation under the locally finite measure gives $F_\varepsilon^\rho \in C^\infty$, and positivity of ρ near zero gives $F_\varepsilon^\rho(y) > 0$ for every y and every $\varepsilon > 0$ on that same event. Compactness gives $\inf_K F_\varepsilon^\rho > 0$, so $H_\varepsilon^\rho = \log F_\varepsilon^\rho$ is smooth and locally integrable.

1.2 Literature review

Caravenna, Sun, and Zygouras constructed the critical two-dimensional SHF as the universal scaling limit of directed polymers in random environment in the critical window [2]. They subsequently showed that the SHF is not a Gaussian multiplicative chaos in the sense of being the exponential of a generalized Gaussian field [3]. This makes a logarithmic field associated with the SHF substantially different from simply taking the logarithm of a pre-existing Gaussian multiplicative-chaos representation.

Tsai gave an axiomatic characterization of the SHF by moments and proved a construction from the noise-mollified stochastic heat equation [5]. Clark and Tsai later identified a conditional GMC structure for SHF polymer measures and, among other consequences, proved strict positivity for SHF tested against general nonnegative functions [7]. Nakashima proved quantitative lower-tail bounds for masses of balls and deduced log-integrability and strict positivity in that setting [8]. These works provide several forms of positivity and lower-tail information, but do not directly give compactness of the full spatially regularized logarithmic field in a distribution topology.

The multiscale starting point of the present proof is Gu and Tsai's analysis of log–log fluctuations [6]. We use their reset/decoupling algebra and quote two local estimates: the single-chain L^2 bound and the one-block high-moment bound. The all-length interval-gas estimates, denominator taming, whole-space positive-part control, and distribution-level compactness argument are developed here. For the lower tail we use Chen's Gaussian convexity theorem [4]; for the final passage from scalar tightness to strong-dual tightness we use Fouque's extension of Mitoma's criterion [9]. Appendix A collects the external results in the precise forms used below.

1.3 Outline of proof

Main challenge. The proof is based on the multiscale framework developed by Gu–Tsai [6]. For nonnegative deterministic profiles f, h and a locally finite measure kernel $K(\mathrm{d}x, \mathrm{d}z)$, write

$$f \triangleleft K \triangleright h := \iint f(x)h(z)\, K(\mathrm{d}x, \mathrm{d}z).$$

When the terminal profile is 1, any moment identity originally stated for a compactly supported terminal test is applied first to nonnegative cutoffs $\chi_R \uparrow 1$ and then passed to the limit by positivity and monotone convergence. At geometric times $\tau_k = b^{2k}$, one compares the mass across a long time window,

$$Y_{m,n}(y) = p_{\tau_n}(\cdot - y) \triangleleft Z^{\theta}_{\tau_n,\tau_m} \triangleright 1,$$

with the product of the corresponding one-block masses

$$G_k(y) = p_{\tau_{k+1}}(\cdot - y) \triangleleft Z^{\theta}_{\tau_{k+1},\tau_k} \triangleright 1$$

The resulting multiscale error is

$$\Lambda_{m,n}(y) = \log Y_{m,n}(y) - \sum_{k=m}^{n-1} \log G_k(y) = \log \frac{Y_{m,n}(y)}{\prod_{k=m}^{n-1} G_k(y)}$$

Thus the main analytic problem is to show that the interaction between different scales, measured by $\Lambda_{m,n}$, remains uniformly small as the number of scales grows.

The principal difficulty is that the two signs of $\Lambda_{m,n}$ require rather different arguments. For the positive part, the Gu–Tsai reset decomposition turns the ratio $Y_{m,n}/\prod_{k=m}^{n-1} G_k$ into a signed disjoint-interval expansion. However, its activities contain the random reciprocals G_k^{-1}, so rare small values of a one-block mass can destroy uniform summability even when the underlying chain estimates are small. For the negative part, there is a more fundamental obstruction: the signed-chain expansion gives an effective upper bound for the logarithmic error, but it does not by itself prevent $Y_{m,n}$ from being anomalously small relative to the product of the one-block masses.

Main idea. The central idea of our proof is to treat the positive and negative sides of the multiscale error $\Lambda_{m,n}$ by two different, complementary mechanisms.

To control the positive side, we introduce a *denominator-taming* procedure: exceptionally small one-scale masses are raised to a fixed positive threshold. Crucially, this is achieved by adding a positive term to the underlying evolution, rather than merely truncating a denominator $G_k(y)$ in an estimate. The added term modifies the reference evolution while preserving the signed corrections responsible for interactions between scales. We thus obtain a larger, better-conditioned comparison model to which the multiscale estimates stemming from Gu–Tsai [6] can be applied. This gives control of positive deviations uniformly over arbitrarily many scales, without discarding exceptional spatial points.

For the negative side, we take a different route. Instead of proving a lower-tail estimate for the multiscale error, we only seek a lower bound on its expectation. Chen's Gaussian convexity theorem [4], transferred to SHF through Tsai's Gaussian approximations [5], allows us to turn a second-moment bound into a lower bound on the expected logarithm of a positive SHF mass. This free-energy bound matches the leading drift of the single-scale logarithms. Together with the positive-side estimate, it yields the desired two-sided L^1 control via the elementary identity

$$\mathbb{E}|X| = 2\mathbb{E}[X_+] - \mathbb{E}[X].$$

The key insight is that a positivity-preserving upper comparison and a convexity-based bound on the mean supply exactly the complementary information needed for multiscale decoupling. This, in turn, is the main probabilistic input for establishing compactness of the spatially regularized logarithmic fields as distributions.

1.4 Outline of the paper

Section 2 develops the reset decomposition, the exact interval-gas identity, and the multiscale signed-chain summation. Section 3 introduces denominator taming and proves whole-space positive-part control. Section 4 combines Gaussian convexity, exact second moments, and the argument that closes the negative-spike obstruction. Section 5 proves the spatially tested fine-window estimate and transfers it to fixed times, general initial data, and each fixed mollifier. Section 6 upgrades the scalar estimate to tightness in the strong dual. Appendix A records the external theorems used in the proof.

2 Full multiscale signed-chain argument

For nonnegative deterministic profiles f, h and a locally finite measure kernel $K(\mathrm{d}x, \mathrm{d}z)$, write

$$f \triangleleft K \triangleright h := \iint f(x)h(z)\, K(\mathrm{d}x, \mathrm{d}z).$$

When the terminal profile is 1, any moment identity originally stated for a compactly supported terminal test is applied first to nonnegative cutoffs $\chi_R \uparrow 1$ and then passed to the limit by positivity and monotone convergence.

Fix $b > 0$ sufficiently small for the signed-chain summations below, and set

$$L = \log b^{-1}, \qquad \tau_k = b^{2k}.$$

All quantities in an aligned window are constructed from the same SHF:

$$G_k(y) = p_{\tau_{k+1}}(\cdot - y) \triangleleft Z^{\theta}_{\tau_{k+1},\tau_k} \triangleright 1, \tag{3}$$

$$Y_{m,n}(y) = p_{\tau_n}(\cdot - y) \triangleleft Z^{\theta}_{\tau_n,\tau_m} \triangleright 1, \tag{4}$$

$$\Lambda_{m,n}(y) = \log Y_{m,n}(y) - \sum_{k=m}^{n-1} \log G_k(y). \tag{5}$$

The G_k are independent random spatial fields, and $\mathbb{E}G_k = \mathbb{E}Y_{m,n} = 1$. For any functions f, g on $\mathbb{R}^2$ and any measures K, L on $\mathbb{R}^4$, define the composition mollified by the Gaussian bridge by

$$f \triangleleft K \bullet L \triangleright g := \int_{\mathbb{R}^8} f(x)K(dx, du)p_{\varepsilon}(u - v)L(dv, dz)g(y).$$

Then we have

$$Y_{m,n}(y) = p_{\tau_n}(\cdot - y) \triangleleft \mathcal{Z}_{n-1} \bullet \mathcal{Z}_{n-2} \bullet \cdots \bullet \mathcal{Z}_m \triangleright 1$$

The external microscopic inputs for this section are listed in Appendix A. In particular, we quote only the local Gu–Tsai estimates; all multiscale consequences needed below are proved here.

The next theorem is the main output of the multiscale analysis. It quantifies how well a long aligned SHF mass is approximated, on the logarithmic scale, by the product of its independent one-block masses; later sections use this as the error estimate in the spatially tested decomposition.

Theorem 2.1 (Two-sided spatial decoupling)**.** *For every integer $m \geq 1$, the free-energy estimate*

$$\sup_{n>m} [-\mathbb{E}\Lambda_{m,n}(y)]_+ \leq Cm^{-1/2} \tag{6}$$

holds uniformly in y. Consequently, for every compact K,

$$\boxed{\sup_{n>m\geq M} \mathbb{E} \int_K |\Lambda_{m,n}(y)| \, \mathrm{d}y \leq C|K|M^{-1/2}.} \tag{7}$$

This section supplies the complete multiscale signed-chain argument used in the proof. We quote only the single-chain estimate of Gu–Tsai and prove every multiscale consequence used later.

2.1 Reset operators and the exact interval-gas expansion

The reset/decoupling algebra in this subsection comes from Gu–Tsai [6]. We restate it in the present reversed scale index and use the expository names *reset kernel*, *reset error*, and *interval gas*.

Write

$$\mathcal{Z}_k = Z^{\theta}_{\tau_{k+1},\tau_k}, \qquad p_{k,y} = p_{\tau_k}(\cdot - y).$$

For a positive kernel K we use $K \triangleright 1$ for its right mass. Define the rank-one **reset kernel** and **reset error** by

$$R_{k,y} = (\mathcal{Z}_k \triangleright 1) \otimes p_{k,y}, \qquad D_{k,y} = \mathcal{Z}_k - R_{k,y}. \tag{8}$$

Note that

$$R_{k,y}(\,\mathrm{d}x, \,\mathrm{d}z) = \mathcal{Z}_k \triangleright 1(\,\mathrm{d}x)\, p_{k,y}(z)\,\mathrm{d}z.$$

Consequently, once a composition hits $R_{k,y}$, all dependence on the earlier left profile is compressed into the scalar $f \triangleleft \mu_k = f \triangleleft \mathcal{Z}_k \triangleright 1$, while the continuation restarts from the deterministic profile $p_{k,y}$. This is the precise sense in which $R_{k,y}$ "resets" the chain.

Moreover, the reset error has a fundamental cancellation

$$D_{k,y} \triangleright 1 = 0, \tag{9}$$

and, by the mean heat-kernel identity and the heat semigroup,

$$p_{k+1,y} \triangleleft \mathbb{E}D_{k,y} = p_{k+1,y} \triangleleft \mathbb{E}(\mathcal{Z}_k - R_{k,y}) = p_{k,y} - p_{k,y} = 0. \tag{10}$$

where we used the fact that $\mathbb{E}\mathcal{Z}_k$ is the heat kernel from τ_{k+1} to τ_k and $\mathbb{E}(\mathcal{Z}_k \triangleright 1) = 1$.

Using $\mathcal{Z}_k = R_{k,y} + D_{k,y}$, the mass $Y_{m,n}(y)$ can be written as

$$\begin{aligned} Y_{m,n}(y) &= p_{\tau_n}(\cdot - y) \triangleleft \mathcal{Z}_{n-1} \bullet \mathcal{Z}_{n-2} \bullet \cdots \bullet \mathcal{Z}_m \triangleright 1 &(11)\\ &= p_{\tau_n}(\cdot - y) \triangleleft (R_{n-1,y} + D_{n-1,y}) \bullet \cdots \bullet (R_{m,y} + D_{m,y}) \triangleright 1. &(12) \end{aligned}$$

Formally, the last line can be expanded into a sum of sequences consisting of $R_{k,y}$ and $D_{k,y}$; however, the compositions here involve Gaussian bridges as mollifiers, so the existence of such an expansion needs to be justified, which is given in Lemma 3.3, Step 3. Since the error $D_{k,y}$ has zero right mass, a lone terminal D contributes nothing, for instance,

$$Y_{n-1,n}(y) = p_{\tau_n}(\cdot - y) \triangleleft \mathcal{Z}_{n-1} \triangleright 1 = p_{\tau_n}(\cdot - y) \triangleleft R_{n-1} \triangleright 1.$$

Hence nontrivial error contributions must therefore come in composition of at least two kernels $\mathcal{Z}_k$, that is, $n - m > 1$ for the indices of $Y_{m,n}$.

Now, for $J = [j, j+h-1]$, $h \geq 2$, define

$$A_J(y) = p_{j+h,y} \triangleleft D_{j+h-1,y} \bullet \cdots \bullet D_{j+1,y} \bullet \mathcal{Z}_j \triangleright 1, \qquad a_J(y) = \frac{A_J(y)}{\prod_{k\in J} G_k(y)}. \tag{13}$$

For any set of nonnegative integers (not necessarily an interval) $V \subset \mathbb{N}_{\geq 0}$, define

$$\mathcal{I}(V) := \left\{ I = \{J_1, \ldots, J_{|I|}\} : \begin{array}{l} |I| \geq 1, \quad J_1, \ldots, J_{|I|} \text{ are pairwise disjoint integer intervals} \\ \text{contained in } V, |J_a| \geq 2 \text{ for every } a \end{array} \right\}.$$

Thus I denotes an interval family and J denotes a single interval.

The following identity is the algebraic bridge between SHF compositions and a one-dimensional gas of disjoint error intervals. Its purpose is to isolate every failure of a reset into a local activity a_J that can subsequently be estimated and summed.

Proposition 2.2 (Exact disjoint-interval gas identity)**.** *For every $n > m$,*

$$\boxed{e^{\Lambda_{m,n}(y)} = 1 + \sum_{I\in\mathcal{I}([m,n])} \prod_{J\in I} a_J(y).} \tag{14}$$

The identity is pathwise on the common event on which all displayed masses are finite and positive.

Proof. In each slab write $\mathcal{Z}_k = R_{k,y} + D_{k,y}$. The rank-one factor satisfies, for every left profile f and every continuation kernel K,

$$f \triangleleft R_{k,y} \bullet K \triangleright 1 = \big(f \triangleleft \mathcal{Z}_k \triangleright 1\big)\big(p_{k,y} \triangleleft K \triangleright 1\big). \tag{15}$$

Thus a reset closes the current dependence on all earlier kernels and starts the remaining composition from the deterministic profile $p_{k,y}$.

Formally expand $Y_{m,n}$ as the product of the decompositions $R_k + D_k$ as in (11), scanning from the finest slab toward the coarsest one. A string containing only resets contributes the product $\prod_{k=m}^{n-1} G_k$. Whenever the expansion first chooses a D and later returns to a reset, the intervening consecutive run is forced by (15) to contribute exactly one factor A_J divided by the corresponding product of one-slab masses. The run has length at least two: a D immediately tested against terminal 1 vanishes by (9). Two distinct runs cannot overlap, while any pairwise disjoint family of runs is produced exactly once. Repeated use of (15) therefore yields

$$Y_{m,n}(y) = \prod_{k=m}^{n-1} G_k(y) \left(1 + \sum_{I\in\mathcal{I}([m,n])} \prod_{J\in I} \frac{A_J(y)}{\prod_{k\in J} G_k(y)} \right).$$

Division by the strictly positive product of the G_k gives (14). This is the same algebra as Gu–Tsai's exact decoupling expansion, written in the present reversed scale index. The measure-theoretic justification of the rank-one reset products under Gaussian bridge regularization is supplied in Lemma 3.3, Step 3; that later lemma does not use Proposition 2.2, so there is no circularity. □

Why this is an interval gas. On the right-hand side of (14), every configuration is encoded by a family of pairwise disjoint integer intervals. An interval J is a "polymer" carrying signed activity a_J, and disjointness is the hard-core interaction. The terminology is useful because all subsequent logarithmic estimates are deterministic estimates for this one-dimensional hard-core gas; the stochastic analysis enters only through bounds on the random activities.

2.2 The sole signed-chain estimate quoted from Gu–Tsai

The following is the quoted from Gu–Tsai. This imported local estimate is the microscopic quantitative input for the whole multiscale argument. It controls one error interval of a fixed length; the rest of Sections 2–3 is devoted to summing such intervals uniformly over all lengths and cutoffs.

Lemma 2.3 (Single-chain L^2 estimate; Gu–Tsai)**.** *Recall that $A_J(y)$ is defined in* (13)*. There is $C_0 < \infty$ such that, uniformly in y, j and $h \geq 2$,*

$$\mathbb{E}|A_{[j,j+h-1]}(y)|^2 \leq \frac{C_0^h}{L^{h-1} j^2}, \qquad L = \log b^{-1}. \tag{16}$$

Moreover, for every fixed even r, there is $C_r < \infty$ such that

$$\mathbb{E}|G_k(y) - 1|^r \leq C_r k^{-r/2}, \qquad k \geq 1, \tag{17}$$

where C_r may depend on b.

Proof. For the chain bound choose an auxiliary integer $N_* > j + h$ and put $\varepsilon = b^{N_*}$. Gu–Tsai use the geometric clock

$$t_i = \varepsilon^2 b^{-2i} = b^{2(N_* - i)}.$$

Under the reversal $i = N_* - k$ this is exactly our clock $\tau_k = b^{2k}$. Our interval $[j, j + h - 1]$ becomes a Gu–Tsai interval of the same length h, and their distance-to-the-terminal-index factor $N_* - n'$ is exactly j. Their signed chain $pDZI$ is, after this substitution and a spatial translation by y, the quantity in (13). Corollary 3.5, formula (3.40), of [6] is therefore precisely (16). Translation invariance removes y.

Likewise Corollary B.3, formula (B.22), of [6], after the same reversal of the geometric index, gives, for each fixed even r, constants $C_r < \infty$ and $k_r < \infty$ such that

$$\mathbb{E}|G_k(y) - 1|^r \leq C_r k^{-r/2}, \qquad k \geq k_r. \tag{18}$$

Indeed, their restriction $i \in [1, N - c(r)]$ becomes $k = N - i \geq c(r)$. This is the only range supplied directly by that corollary. For the finitely many indices $1 \leq k < k_r$, finiteness of the fixed-time r-th moments follows from the all-order SHF moment-measure formula (Proposition 1.7 and Corollary 1.8 of [5]), approximating the terminal test 1 monotonically by compactly supported cutoffs and using positivity. Since there are only finitely many such k, enlarging C_r gives (17) for all $k \geq 1$. □

The $r = 4$ case will be used repeatedly below:

$$\mathbb{E}|G_k(y) - 1|^4 \leq Ck^{-2}. \tag{19}$$

In particular, the second- and fourth-moment bounds give

$$\|G_k - 1\|_2 \leq Ck^{-1/2}, \qquad \mathbb{P}(|G_k - 1| > 1/2) \leq Ck^{-2}. \tag{20}$$

2.3 A deterministic logarithm lemma for a signed interval gas

The following perturbative lemma below turns small local interval activities into a second-order control of the logarithm of the gas partition function. It is the basic deterministic estimate behind the later random multiscale bounds.

Lemma 2.4 (Small-activity signed interval gas)**.** *Let $V = [m, n] \cap \mathbb{Z}$ and let $a_J \in \mathbb{R}$ be activities indexed by subintervals $J \subset V$. Put*

$$\mathcal{P}(V) = \sum_{I \in \mathcal{I}([m,n])} \prod_{J \in I} a_J, \qquad h_r = \sum_{J \ni r} e^{|J|} |a_J|, \qquad H = \sum_{r \in V} h_r^2.$$

There are universal constants $0 < \eta_0 < 1/2$ and $C_{\mathrm{gas}} < \infty$ such that, if $\sup_r h_r \le \eta_0$, then $\mathcal{P}(V) > 0$ and

$$\boxed{\left| \log \mathcal{P}(V) - \sum_{J \subset V} a_J \right| \le C_{\mathrm{gas}} H.} \tag{21}$$

Proof. Given $0 < \eta_0 < 1/2$, we suppose $\sup_r h_r \le \eta_0$ from now on.

For $r \in V$ let $P_r = \mathcal{P}([m, r])$ be the gas partition function using only intervals contained in $[m, r]$, and set $P_{m-1} = 1$. Decomposing according to the rightmost selected interval gives the exact recursion

$$P_r = P_{r-1} + \sum_{J=[i,r] \subset V} a_J P_{i-1}, \tag{22}$$

where the sum is over all subintervals $J \subset V$ with the right endpoint being r; to put it differently, the sum is over all $J = [i, r]$ as i runs over $i = m, m + 1, \cdots, r$.

As long as the previous P_s are positive, write

$$P_s / P_{s-1} = 1 + x_s.$$

Then (22) becomes

$$x_r = \sum_{J=[i,r] \subset V} a_J \prod_{s=i}^{r-1} (1 + x_s)^{-1}. \tag{23}$$

Note that if all x_s are small, then we roughly have $\log P_r - \log P_{r-1} \approx x_r$ and $x_r \approx \sum_{J=[i,r] \subset V} a_J$, which imply

$$\log \mathcal{P}(V) = \log P_n \approx \sum_r \sum_{J=[i,r] \subset V} a_J = \sum_{J \subset V} a_J.$$

Our goal is to obtain inequality (21), which is exactly about the error of such approximation. So we are going to quantify the above approximations in the rest of the proof.

We prove inductively that $|x_s| \le h_s \le \eta_0$. For the initial case $r = m$, we have

$$x_m = a_{[m,m]} \le e |a_{[m,m]}| \le h_m \le \eta_0,$$

as desired. For the inductive step, if this holds before r, then

$$\prod_{s=i}^{r-1} |1 + x_s|^{-1} \le \exp\left(2 \sum_{s=i}^{r-1} |x_s| \right) \le e^{2\eta_0 |J|} \le e^{|J|},$$

where we used the hypothesis $|x_r| \le \eta_0 < 1/2$. So $|x_r| \le h_r$. Thus every $P_r > 0$.

For $|x_s| \le \eta_0$, the elementary product estimate

$$\left| \prod_{s=i}^{r-1} (1 + x_s)^{-1} - 1 \right| \le C e^{|J|/2} \sum_{s=i}^{r-1} |x_s|$$

holds after decreasing η_0 if needed. One way to see it is to write

$$\prod_{s=i}^{r-1}(1+x_s)^{-1}-1=\sum_{t=i}^{r-1}\big((1+x_t)^{-1}-1\big)\prod_{s=i}^{t-1}(1+x_s)^{-1}.$$

For $|x|\le\eta_0<1/2$, one has $|(1+x)^{-1}-1|\le 2|x|$ and $|(1+x)^{-1}|\le e^{2|x|}\le e^{2\eta_0}\le e$. Hence the tth term is at most $2|x_t|e^{2\eta_0|J|}$, which gives the above bound. Therefore

$$\begin{aligned}\sum_{r\in V}\left|x_r-\sum_{J=[i,r]\subset V}a_J\right|=\sum_{r\in V}\left|\sum_{J=[i,r]\subset V}a_J\left[\prod_{s=i}^{r-1}(1+x_s)^{-1}-1\right]\right|&\le C\sum_{J\subset V}e^{|J|/2}|a_J|\sum_{s\in J}h_s\\&\le C\sum_{s\in V}h_s\sum_{J\ni s}e^{|J|}|a_J|=C\sum_{s\in V}h_s^2.\end{aligned}\tag{24}$$

Also $|\log(1+x)-x|\le Cx^2$ on $|x|\le\eta_0$, whence

$$\left|\log P_n-\sum_{r\in V}x_r\right|=\left|\sum_{r\in V}(\log(1+x_r)-x_r)\right|\le C\sum_{r\in V}h_r^2.\tag{25}$$

Finally, we have

$$\sum_{r\in V}\sum_{J=[i,r]\subset V}a_J=\sum_{J\subset V}a_J.$$

Combining (24) and (25) proves (21). □

The next one-sided form removes the smallness hypothesis $\sup_r h_r\le\eta_0$. It is the deterministic input needed to control positive logarithmic spikes everywhere in space. Note that the below lemma removes the smallness assumption at the price of a one-sided estimate, but it's sufficient for the later purpose.

Lemma 2.5 (Nonperturbative upper bound)**.** *With the same notation, assume only that the finite signed gas satisfies $\mathcal{P}(V)>0$. Then*

$$\boxed{\log\mathcal{P}(V)\le\sum_J a_J+C_\star\sum_r h_r^2,}\tag{26}$$

where $C_\star$ is universal.

Proof. Let $B=\{r:h_r>\eta_0\}$ and call an interval bad if it meets B, good otherwise. Define the configuration of good/bad subintervals of V by

$$\begin{aligned}\mathcal{I}_g(V)&:=\{I\in\mathcal{I}(V):J\in I\text{ implies that }J\text{ is good}\},\\\mathcal{I}_b(V)&:=\{I\in\mathcal{I}(V):J\in I\text{ implies that }J\text{ is bad}\}.\end{aligned}$$

Define the good-only gas $\mathcal{P}_g$ by

$$\mathcal{P}_g(V):=\sum_{I\in\mathcal{I}_g(V)}\prod_{J\in I}a_J$$

Note that $\mathcal{P}_g$ satisfies Lemma 2.4 even if the sum on the right-hand side is over the family of subintervals in $\mathcal{I}_g(V)$ instead of $\mathcal{I}(V)$. It's because we can write

$$\mathcal{P}_g(V) = \sum_{I \in \mathcal{I}_g(V)} \prod_{J \in I} a_J = \sum_{I \in \mathcal{I}(V)} \prod_{J \in I} a'_J$$

where we define the activity on the right-hand side by

$$a'_J := \begin{cases} a_J \text{ if } J \text{ is good} \\ 0 \text{ if } J \text{ is bad.} \end{cases}$$

If $E \subset V$, then $\mathcal{P}_g(V \setminus E)$ satisfies Lemma 2.4 similarly. Two applications of (21) give

$$\begin{aligned} \log \frac{\mathcal{P}_g(V \setminus E)}{\mathcal{P}_g(V)} &\le \sum_{\substack{J \text{ is good} \\ J \cap E \neq \varnothing}} |a_J| + 2C_{\text{gas}}H \\ &\le \sum_{r \in E} \sum_{\substack{J \text{ is good} \\ J \ni r}} |a_J| + 2C_{\text{gas}}H \\ &\le \eta_0 |E| + 2C_{\text{gas}}H. \end{aligned} \tag{27}$$

Now we expand the full gas according to its selected bad intervals. Given $I \in \mathcal{I}(V)$, I can be decomposed into the disjoint union $I = I_g \sqcup I_b$ where I_g contains only good intervals and I_b contains only bad intervals. Conversely, if we fix a family of bad intervals I_b, then we can choose the complementary family of good intervals I_g from $V \setminus \bigcup I_b$ and combine them to form a family of intervals. This gives the identity

$$\begin{aligned} \mathcal{P}(V) &= \sum_{I_b \in \mathcal{I}_b(V)} \prod_{J \in I_b} a_J \sum_{I \in \mathcal{I}_g(V \setminus \bigcup I_b)} \prod_{K \in I} a_K \\ &= \sum_{I_b \in \mathcal{I}_b(V)} \prod_{J \in I_b} a_J \cdot \mathcal{P}_g(V \setminus \bigcup I_b). \end{aligned} \tag{28}$$

Denote the collection of good/bad subintervals of V by

$$\begin{aligned} \mathcal{G} &= \{J \subset V : J \text{ is a good interval}\}, \\ \mathcal{B} &= \{J \subset V : J \text{ is a bad interval}\}. \end{aligned}$$

Applying (27) to (28), we obtain

$$\begin{aligned} \mathcal{P}(V) &\le \mathcal{P}_g(V) e^{2C_{\text{gas}}H} \sum_{I_b \in \mathcal{I}_b(V)} \prod_{J \in I_b} e^{\eta_0 |J|} |a_J| \\ &\le \mathcal{P}_g(V) e^{2C_{\text{gas}}H} \sum_{I_b \subset \mathcal{B}} \prod_{J \in I_b} e^{\eta_0 |J|} |a_J| \\ &= \mathcal{P}_g(V) e^{2C_{\text{gas}}H} \prod_{J \in \mathcal{B}} (1 + e^{\eta_0 |J|} |a_J|) \\ &\le \mathcal{P}_g(V) \exp\left(2C_{\text{gas}}H + \sum_{J \in \mathcal{B}} e^{\eta_0 |J|} |a_J| \right), \end{aligned}$$

where we used (27) in the first inequality, and we relax the condition that I_b must contain bad subintervals in the second inequality. Now

$$\log \mathcal{P}_g(V) \le \sum_{J\in\mathcal{G}} a_J + C_{\mathrm{gas}} H.$$

Since

$$\sum_{J\in\mathcal{G}} a_J = \sum_{J\subset V} a_J - \sum_{J\in\mathcal{B}} a_J,$$

we obtain

$$\log \mathcal{P}(V) \le \sum_{J\subset V} a_J + 3C_{\mathrm{gas}} H + \sum_{J\in\mathcal{B}} \left(|a_J| + e^{\eta_0|J|}|a_J| \right).$$

Because $\eta_0 < 1$ and $|J| \ge 1$,

$$|a_J| + e^{\eta_0|J|}|a_J| \le 2e^{|J|}|a_J|,$$

so restoring the bad linear terms costs at most

$$C \sum_{J\in\mathcal{B}} e^{|J|}|a_J|.$$

It remains to control this sum by H. By definition, every bad interval $J \in \mathcal{B}$ meets

$$B = \{r \in V : h_r > \eta_0\}.$$

Hence each bad interval contains at least one point $r \in B$, and therefore, allowing overcounting,

$$\sum_{J\in\mathcal{B}} e^{|J|}|a_J| \le \sum_{r\in B} \sum_{J\ni r} e^{|J|}|a_J| = \sum_{r\in B} h_r.$$

For every $r \in B$ we have $h_r > \eta_0$, so

$$h_r \le \eta_0^{-1} h_r^2.$$

Consequently,

$$\sum_{J\in\mathcal{B}} e^{|J|}|a_J| \le \sum_{r\in B} h_r \le \eta_0^{-1} \sum_{r\in B} h_r^2 \le \eta_0^{-1} H.$$

Combining the preceding estimates and absorbing all universal constants into $C_\star$, we obtain

$$\log \mathcal{P}(V) \le \sum_{J\subset V} a_J + C_\star H,$$

which is exactly (26). □

3 Denominator taming and positive-part control

We now prove the denominator-taming argument that controls positive logarithmic spikes on the whole space. The device replaces every potentially small G_k by $g_k^* = G_k \vee \frac{1}{2}$, so the reciprocal is globally bounded by 2. Through the positive-kernel realization below, it also gives the monotonicity $Y \le Y^*$ and therefore a whole-space one-sided upper bound.

Set

$$g_k^* = G_k \vee \tfrac{1}{2}, \qquad a_J^* = \frac{A_J}{\prod_{k\in J} g_k^*}. \tag{29}$$

Clipping the denominators is useful only if the clipping itself is negligible. The following lemma shows that the accumulated logarithmic cost of replacing G_k by $G_k \vee \frac{1}{2}$ decays faster than any polynomial in the starting scale.

Lemma 3.1. *For every fixed $p \geq 1$ and $A > 0$,*

$$\sup_y \left\| \sum_{k \geq M} \log(g_k^*/G_k) \right\|_p \leq C_{p,A} M^{-A}. \tag{30}$$

Proof. Fix $p \geq 1$ and $A > 0$. Note that $\log(g_k^*/G_k) = [-\log(2G_k)]_+$. Thus $\log(g_k^*/G_k) = 0$ whenever $G_k \geq \frac{1}{2}$, while on the event $\{\log(g_k^*/G_k) > 0\} = \{G_k < \frac{1}{2}\}$ we have

$$|\log G_k| \geq \log 2$$

and

$$0 < \log(g_k^*/G_k) = -\log 2 - \log G_k \leq |\log G_k|.$$

Choose an exponent $r > p$, to be specified below. Since $|\log G_k| \geq \log 2$ on $\{\log(g_k^*/G_k) > 0\}$, and $p - r < 0$, we have there

$$|\log G_k|^{p-r} \leq (\log 2)^{p-r}.$$

Consequently,

$$|\log(g_k^*/G_k)|^p \leq \mathbf{1}_{\log(g_k^*/G_k)>0} |\log G_k|^p = \mathbf{1}_{\log(g_k^*/G_k)>0} |\log G_k|^r |\log G_k|^{p-r} \leq (\log 2)^{p-r} |\log G_k|^r.$$

Taking expectations and using Lemma 4.5, which gives

$$\|\log G_k\|_r \leq C_r k^{-1/2},$$

we obtain

$$\mathbb{E}|\log(g_k^*/G_k)|^p \leq (\log 2)^{p-r} \mathbb{E}|\log G_k|^r \leq C_r k^{-r/2}.$$

Hence

$$\|\log(g_k^*/G_k)\|_p \leq C_{p,r} k^{-r/(2p)}.$$

Now apply Minkowski's inequality:

$$\left\| \sum_{k \geq M} \log(g_k^*/G_k) \right\|_p \leq \sum_{k \geq M} \|\log(g_k^*/G_k)\|_p \leq C_{p,r} \sum_{k \geq M} k^{-r/(2p)}.$$

Choose r sufficiently large so that

$$\frac{r}{2p} > A + 1.$$

Then the tail of the p-series satisfies

$$\sum_{k \geq M} k^{-r/(2p)} \leq C_{p,r,A} M^{1-r/(2p)} \leq C_{p,r,A} M^{-A}.$$

Therefore

$$\sup_y \left\| \sum_{k \geq M} \log(g_k^*/G_k) \right\|_p \leq C_{p,A} M^{-A}.$$

The estimates in Lemma 4.5 are uniform in y, so the same is true of the final bound.

□

The following lemma is the summation engine for the tamed gas. It converts the single-chain estimate into bounds uniform over interval length, starting point, and terminal cutoff, including the maximal estimate needed below.

Lemma 3.2 (Tamed activity estimates and their all-length sums)**.** *For b sufficiently small there is some $q < e^{-2}$ such that*

$$\|a^*_{[j,j+h-1]}\|_2 \le Cq^h/j, \qquad |\mathbb{E}a^*_{[j,j+h-1]}| \le Cq^h/j^{3/2}. \tag{31}$$

Set

$$S^*_{m,n} = \sum_{\substack{J\subset[m,n-1]\\ |J|\ge 2}} a^*_J, \qquad h^*_{r,M} = \sum_{\substack{J\subset[M,\infty)\\ r\in J}} e^{|J|}|a^*_J|, \qquad H^*_M = \sum_{r\ge M}(h^*_{r,M})^2.$$

Then

$$\mathbb{E}\sup_{n>m\ge M}|S^*_{m,n}|^2 \le C/M, \qquad \mathbb{E}H^*_M \le C/M. \tag{32}$$

Proof. Write $J = [j, j+h-1]$ and $r = j+h-1$. Since $d_k = (G_k \vee \frac{1}{2})^{-1} \le 2$, the single-chain estimate (16) gives

$$\|a^*_J\|_2 \le 2^h\|A_J\|_2 \tag{33}$$

$$\le \frac{\sqrt{L}}{j}\left(2\sqrt{C_0/L}\right)^h. \tag{34}$$

Choose $L = \log b^{-1}$ sufficiently large that

$$q := 2\sqrt{C_0/L} < e^{-2}.$$

Since L is then fixed, the factor $\sqrt{L}$ is absorbed into the constant, proving the first bound in (31).

For the mean, condition on all slabs except the rth one. If the factor $(g^*_r)^{-1}$ is replaced by 1, the conditional expectation of the resulting term vanishes by (10). Hence

$$\mathbb{E}a^*_J = \mathbb{E}\left[A_J\left(\frac{1}{g^*_r}-1\right)\prod_{k\in J\setminus\{r\}}\frac{1}{g^*_k}\right]. \tag{35}$$

For every $x > 0$,

$$\left|(x\vee\tfrac{1}{2})^{-1}-1\right| \le 2|x-1| :$$

indeed, for $x \ge \frac{1}{2}$ this follows from $|x^{-1}-1| = |x-1|/x \le 2|x-1|$, while for $x < \frac{1}{2}$ the left side equals 1 and the right side is at least 1. Therefore, by (20),

$$\left\|\frac{1}{g^*_r}-1\right\|_2 \le 2\|G_r-1\|_2 \le Cr^{-1/2}.$$

Using $(g^*_k)^{-1} \le 2$, Cauchy–Schwarz, and (16) in (35),

$$|\mathbb{E}a^*_J| \le 2^{h-1}\|A_J\|_2\|(g^*_r)^{-1}-1\|_2$$
$$\le \frac{Cq^h}{j\sqrt{r}} \le \frac{Cq^h}{j^{3/2}},$$

which proves the second bound in (31).

We now prove the first estimate in (32). For $h \geq 2$ set

$$a^*_{j,h} := a^*_{[j,j+h-1]}, \qquad X_{j,h} := a^*_{j,h} - \mathbb{E}a^*_{j,h}.$$

For fixed h, the random variable $X_{j,h}$ depends only on the slabs $j, \dots, j+h-1$. Thus, for each residue class $\rho \pmod h$, the variables

$$\{X_{j,h} : j \geq M,\ j \equiv \rho \pmod h\}$$

are independent and centered. Enumerate the corresponding starting indices as $j_0 < j_1 < \cdots$. With respect to the filtration generated by these variables,

$$M_N := \sum_{\ell=0}^{N} X_{j_\ell,h}$$

is an L^2 martingale, and independence gives

$$\mathbb{E}|M_N|^2 = \sum_{\ell=0}^{N} \mathbb{E}|X_{j_\ell,h}|^2.$$

Doob's L^2 maximal inequality yields

$$\mathbb{E} \max_{0\leq n\leq N} |M_n|^2 \leq 4 \sum_{\ell=0}^{N} \mathbb{E}|X_{j_\ell,h}|^2.$$

Every consecutive sum in this residue class is a difference of two prefix sums, so

$$\sup_{0\leq a\leq b\leq N} \left| \sum_{\ell=a}^{b} X_{j_\ell,h} \right| \leq 2 \max_{0\leq n\leq N} |M_n|.$$

Letting $N \to \infty$ and using (31),

$$\left\| \sup_{M\leq p\leq q'} \left| \sum_{\substack{p\leq j\leq q' \\ j\equiv\rho \bmod h}} X_{j,h} \right| \right\|_2 \leq Cq^h \left(\sum_{\substack{j\geq M \\ j\equiv\rho \bmod h}} j^{-2} \right)^{1/2}. \tag{36}$$

Summing over the h residue classes and applying Cauchy–Schwarz in ρ gives

$$\begin{aligned} \left\| \sup_{M\leq p\leq q'} \left| \sum_{j=p}^{q'} X_{j,h} \right| \right\|_2 &\leq Cq^h \sum_{\rho=0}^{h-1} \left(\sum_{\substack{j\geq M \\ j\equiv\rho \bmod h}} j^{-2} \right)^{1/2} \\ &\leq C\sqrt{h}\, q^h \left(\sum_{j\geq M} j^{-2} \right)^{1/2} \\ &\leq C\sqrt{h}\, q^h M^{-1/2}. \end{aligned}$$

The deterministic means satisfy, by (31),

$$\sup_{M\le p\le q'}\left|\sum_{j=p}^{q'}\mathbb{E}a^*_{j,h}\right|\le Cq^h\sum_{j\ge M}j^{-3/2}\le Cq^hM^{-1/2}.$$

For fixed h, the length-h contribution to any $S^*_{m,n}$ is a consecutive sum in the starting index j. Hence the preceding two estimates and Minkowski's inequality over $h\ge 2$ give

$$\left\|\sup_{n>m\ge M}|S^*_{m,n}|\right\|_2\le CM^{-1/2}\sum_{h\ge 2}(1+\sqrt{h})q^h\le CM^{-1/2},$$

because $q<1$. Squaring proves

$$\mathbb{E}\sup_{n>m\ge M}|S^*_{m,n}|^2\le C/M.$$

It remains to prove the quadratic incidence bound. By Minkowski's inequality and the first estimate in (31), for every site $r\ge M$,

$$\|h^*_{r,M}\|_2\le\sum_{h\ge 2}e^h\sum_{\substack{j\ge M\\ j\le r\le j+h-1}}\|a^*_{j,h}\|_2 \tag{37}$$

$$\le C\sum_{h\ge 2}(eq)^h\sum_{\substack{j\ge M\\ j\le r\le j+h-1}}\frac{1}{j}. \tag{38}$$

Let

$$f_j=j^{-1}\mathbf{1}_{\{j\ge M\}},\qquad u_h=\mathbf{1}_{\{0,1,\dots,h-1\}}.$$

Up to the harmless choice of convolution orientation, the inner sum in (37) is $(f*u_h)(r)$. Therefore Minkowski in ℓ^2 followed by Young's convolution inequality gives

$$\begin{aligned}(\mathbb{E}H^*_M)^{1/2}&=\left(\sum_{r\ge M}\|h^*_{r,M}\|_2^2\right)^{1/2}\\&\le C\sum_{h\ge 2}(eq)^h\|f*u_h\|_{\ell^2}\\&\le C\sum_{h\ge 2}(eq)^h\|f\|_{\ell^2}\|u_h\|_{\ell^1}\\&=C\sum_{h\ge 2}h(eq)^h\|f\|_{\ell^2}\\&\le CM^{-1/2}.\end{aligned}$$

Here $eq<1$ because $q<e^{-2}$, and $\|f\|^2_{\ell^2}=\sum_{j\ge M}j^{-2}\le C/M$. Squaring completes the proof of the second estimate in (32). □

The next lemma gives denominator taming a genuine positive-kernel realization. Its purpose is twofold: it justifies the formal reset expansion under Gaussian bridge regularization and provides the monotonicity $Y\le Y^*$ needed for the nonperturbative upper bound.

Lemma 3.3 (Positive-kernel realization of denominator taming)**.** *For fixed y define*

$$\mathcal{Z}_k^* = \mathcal{Z}_k + (g_k^* - G_k)\, 1 \otimes p_{k,y}. \tag{39}$$

Then the Gaussian heat-bridge products defining the tested compositions of the $\mathcal{Z}_k^*$ exist in L^1. The kernel $\mathcal{Z}_k^*$ is positive, its one-block mass from $p_{k+1,y}$ is g_k^*, and its reset error is exactly the original one:

$$D_{k,y}^* = D_{k,y}. \tag{40}$$

Moreover, let $Y_{m,n}^*$ denote the resulting composed mass of the modified kernels, that is,

$$Y_{m,n}^*(y) := p_{\tau_n}(\cdot - y) \triangleleft \mathcal{Z}_{n-1}^* \bullet \mathcal{Z}_{n-2}^* \bullet \cdots \bullet \mathcal{Z}_m^* \triangleright 1. \tag{41}$$

Then

$$0 < Y_{m,n} \le Y_{m,n}^* < \infty, \qquad \frac{Y_{m,n}^*}{\prod_{k=m}^{n-1} g_k^*} = 1 + \sum_{I \in \mathcal{I}([m,n])} \prod_{J \in I} a_J^*. \tag{42}$$

For each fixed y, these identities may be chosen on one probability-one event simultaneously for all finite integer intervals.

Before proving this lemma, we explain how we came up with the construction of $\mathcal{Z}_k^*$. The definition of $\mathcal{Z}_k^*$ can be understood as a modification of the reset component that leaves the signed interaction component unchanged.

To see this, introduce the reset projection

$$P_{k,y}(K) := (K \triangleright 1) \otimes p_{k,y},$$

so that

$$R_{k,y} = P_{k,y}(\mathcal{Z}_k), \qquad D_{k,y} = (I - P_{k,y})\mathcal{Z}_k.$$

Since $p_{k,y}$ is a probability density, $P_{k,y}^2 = P_{k,y}$.

We seek a modified kernel $\mathcal{Z}_k^*$ satisfying three properties:

1. Its one-block mass is bounded below by $1/2$.
2. It dominates the original kernel: $\mathcal{Z}_k^* \ge \mathcal{Z}_k$.
3. Its signed reset error is unchanged: $D_{k,y}^* = D_{k,y}$.

The third requirement suggests modifying $\mathcal{Z}_k$ only in the range of $P_{k,y}$. Indeed, if $P_{k,y}(Q) = Q$, then

$$(I - P_{k,y})(\mathcal{Z}_k + Q) = (I - P_{k,y})\mathcal{Z}_k.$$

The simplest positive perturbation of this form is

$$Q = c_k\, 1 \otimes p_{k,y}, \qquad c_k \ge 0.$$

Since the heat profiles have unit mass, adding Q increases the one-block mass from G_k to $G_k + c_k$. Consequently, the choice

$$c_k = \left(\frac{1}{2} - G_k\right)_+ = g_k^* - G_k$$

gives precisely

$$\mathcal{Z}_k^* = \mathcal{Z}_k + (g_k^* - G_k)\, 1 \otimes p_{k,y}.$$

Thus the modification increases the reference, or reset, mass without changing the signed fluctuation $D_{k,y}$. In particular, it regularizes the denominators appearing in the interval activities while preserving the interaction structure. At the same time, positivity of the added rank-one kernel yields the pathwise comparison $\mathcal{Z}_k^* \geq \mathcal{Z}_k$, and hence $Y_{m,n}^* \geq Y_{m,n}$.

This explains why the construction is particularly suited to controlling the positive part of the logarithmic decoupling error: it replaces potentially small one-block normalizations by uniformly positive ones without altering the signed reset errors, while retaining a monotone upper comparison for the full composition.

Proof. Put

$$c_k = g_k^* - G_k = (\tfrac{1}{2} - G_k)_+ \in [0, \tfrac{1}{2}], \qquad Q_{k,y} = 1 \otimes p_{k,y},$$

and, as before,

$$\mu_k := \mathcal{Z}_k \triangleright 1, \qquad R_{k,y} := \mu_k \otimes p_{k,y}, \qquad D_{k,y} := \mathcal{Z}_k - R_{k,y}.$$

All kernels below are measures on $\mathbb{R}^2 \times \mathbb{R}^2$. We use the Gaussian bridge regularization

$$(K \bullet_\epsilon L)(\,\mathrm{d}x,\,\mathrm{d}z) := \int K(\,\mathrm{d}x,\,\mathrm{d}u)\, p_\epsilon(u-v)\, L(\,\mathrm{d}v,\,\mathrm{d}z), \qquad \epsilon > 0. \tag{43}$$

At the interface from slab k to slab $k-1$ we denote the bridge variance by ϵ_k. All finitely many bridge variances are allowed to tend to zero independently.

The approximation argument in this step uses Tsai's compact-test convergence from [5]. The basic composition input is Tsai's compact-test convergence for genuine SHF slabs. We shall first extend it to the noncompact endpoint tests needed here, and then use that extension once and for all to construct every positive word built from $\mathcal{Z}_k, R_{k,y}, Q_{k,y}$.

The Gaussian bridge preserves constants:

$$p_\epsilon * 1 = 1.$$

This identity is what makes all rank-one factorizations below exact.

Step 1: Gaussian incoming profiles and terminal test 1. Fix $s_0 < \cdots < s_N$, put $T = s_N - s_0$, and let

$$K_{\boldsymbol{\epsilon}} = Z^\theta_{s_0,s_1} \bullet_{\epsilon_1} \cdots \bullet_{\epsilon_{N-1}} Z^\theta_{s_{N-1},s_N}.$$

Independence of the time slabs, the mean heat-kernel identity, and Tonelli give

$$\mathbb{E}K_{\boldsymbol{\epsilon}}(\,\mathrm{d}x,\,\mathrm{d}z) = p_{T+\sum_i \epsilon_i}(z-x)\,\mathrm{d}x\,\mathrm{d}z. \tag{44}$$

Let $a > 0$, set $f = p_a(\cdot - y)$, and choose continuous cutoffs $0 \leq \chi_R \leq 1$ with compact support and $\chi_R \uparrow 1$. Put $f_R = f\chi_R$. For fixed R, Tsai's compact-test convergence gives

$$f_R \triangleleft K_{\boldsymbol{\epsilon}} \triangleright \chi_R \longrightarrow f_R \triangleleft Z^\theta_{s_0,s_N} \triangleright \chi_R \qquad \text{in } L^1.$$

We now remove the two cutoffs. Since $K_{\boldsymbol{\epsilon}}$ is positive,

$$0 \leq f \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 - f_R \triangleleft K_{\boldsymbol{\epsilon}} \triangleright \chi_R,$$

and

$$\begin{aligned} &f \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 - f_R \triangleleft K_{\boldsymbol{\epsilon}} \triangleright \chi_R \\ &\quad = f(1-\chi_R) \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 + f\chi_R \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 - \chi_R. \end{aligned}$$

Using (44), $\int p_t(z-x)\,\mathrm{d}z = 1$, and the heat semigroup, we obtain

$$\begin{aligned}&\|f \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 - f_R \triangleleft K_{\boldsymbol{\epsilon}} \triangleright \chi_R\|_1 \\ &\quad\le \int p_a(x-y)(1-\chi_R(x))\,\mathrm{d}x + \int p_{a+T+\sum_i \epsilon_i}(z-y)(1-\chi_R(z))\,\mathrm{d}z. \end{aligned} \tag{45}$$

The right side tends to zero as $R \to \infty$, uniformly whenever $\sum_i \epsilon_i \le 1$: the first term tends to zero by dominated convergence, and the family

$$\{p_{a+t}(\cdot - y) : t \in [T, T+1]\}$$

is uniformly tight.

For the limiting SHF kernel,

$$\mathbb{E} Z^{\theta}_{s_0,s_N}(\,\mathrm{d}x,\,\mathrm{d}z) = p_T(z-x)\,\mathrm{d}x\,\mathrm{d}z,$$

and the same calculation, now with no bridge variances, gives

$$\begin{aligned}&\left\| f \triangleleft Z^{\theta}_{s_0,s_N} \triangleright 1 - f_R \triangleleft Z^{\theta}_{s_0,s_N} \triangleright \chi_R \right\|_1 \\ &\quad\le \int p_a(x-y)(1-\chi_R(x))\,\mathrm{d}x + \int p_{a+T}(z-y)(1-\chi_R(z))\,\mathrm{d}z. \end{aligned} \tag{46}$$

Combining the two cutoff estimates with the compact-test convergence by the triangle inequality yields

$$p_a(\cdot - y) \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 \longrightarrow p_a(\cdot - y) \triangleleft Z^{\theta}_{s_0,s_N} \triangleright 1 \qquad \text{in } L^1. \tag{47}$$

We shall also need a vanishing perturbation of the incoming heat time. If $\eta = \eta(\boldsymbol{\epsilon}) \ge 0$ is deterministic and $\eta \to 0$, then, by positivity,

$$\begin{aligned}&\mathbb{E}\,|p_{a+\eta}(\cdot - y) - p_a(\cdot - y) \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1| \\ &\quad\le \mathbb{E}|p_{a+\eta}(\cdot - y) - p_a(\cdot - y)| \triangleleft K_{\boldsymbol{\epsilon}} \triangleright 1 \\ &\quad= \|p_{a+\eta} - p_a\|_{L^1} \longrightarrow 0. \end{aligned}$$

Hence (47) remains valid with the incoming profile $p_{a+\eta}(\cdot - y)$ on the regularized side and $p_a(\cdot - y)$ on the limiting side.

Step 2: a general positive-word convergence principle. We now prove once and for all the convergence statement that will be used in all subsequent steps.

Consider a finite regularized tested word

$$W_{\boldsymbol{\epsilon}} = f \triangleleft K_d \bullet_{\epsilon_d} K_{d-1} \bullet_{\epsilon_{d-1}} \cdots \bullet_{\epsilon_{a+1}} K_a \triangleright 1, \qquad K_k \in \{\mathcal{Z}_k, R_{k,y}, Q_{k,y}\}, \tag{48}$$

where the incoming profile f is one of the Gaussian profiles arising in our construction, namely $p_{\tau_r+\eta}(\cdot - y)$ with $\eta \ge 0$ deterministic and $\eta \to 0$ along the bridge limit (the case $\eta = 0$ is included). Since every factor in (48) is positive, Tonelli applies.

The two rank-one kernels cut the word as follows. For

$$Q_{k,y} = 1 \otimes p_{k,y},$$

the left leg is the constant 1, and the right leg, after the next bridge, restarts the continuation from

$$p_{k,y} * p_{\epsilon_k} = p_{\tau_k+\epsilon_k}(\cdot - y).$$

Thus a Q_k separates the factors on its two sides and does not retain any genuine SHF slab at the cutting site.

For

$$R_{k,y} = (\mathcal{Z}_k \triangleright 1) \otimes p_{k,y} = \mu_k \otimes p_{k,y},$$

the same rank-one separation occurs, but the left factor contains $\mu_k = \mathcal{Z}_k \triangleright 1$. Consequently slab k is retained as the final slab of the genuine SHF block lying to the left of the cut, while the continuation to the right again restarts from $p_{\tau_k+\epsilon_k}(\cdot - y)$.

Iterating these cuts shows that every positive word (48) factors exactly as

$$W_{\boldsymbol{\epsilon}} = \prod_{\nu=1}^{q} B_{\nu,\boldsymbol{\epsilon}}, \tag{49}$$

where each $B_{\nu,\boldsymbol{\epsilon}}$ is a heat-start, terminal-1 tested composition of *genuine* SHF kernels on one consecutive set of time slabs. Empty pieces contribute the scalar 1. The factors use disjoint time-slab sets and are therefore independent.

By Step 1, including its $p_{a+\eta}$ endpoint extension,

$$B_{\nu,\boldsymbol{\epsilon}} \longrightarrow B_\nu \qquad \text{in } L^1.$$

Every $B_{\nu,\boldsymbol{\epsilon}}$ and B_ν is nonnegative and has mean one. Hence, by telescoping and independence,

$$\left\| \prod_{\nu=1}^{q} B_{\nu,\boldsymbol{\epsilon}} - \prod_{\nu=1}^{q} B_\nu \right\|_1 \le \sum_{\nu=1}^{q} \|B_{\nu,\boldsymbol{\epsilon}} - B_\nu\|_1 \longrightarrow 0. \tag{50}$$

Therefore every finite positive word with factors in $\{\mathcal{Z}_k, R_{k,y}, Q_{k,y}\}$ has a well-defined L^1 Gaussian heat-bridge limit. We shall use this conclusion without repeating the factorization argument.

Step 3: construction of the modified positive composition and monotonicity. The modified kernel

$$\mathcal{Z}_k^* = \mathcal{Z}_k + c_k Q_{k,y}$$

is positive because $\mathcal{Z}_k, Q_{k,y} \ge 0$ and $c_k \ge 0$.

Fix $m < n$, put

$$V = [m, n-1] \cap \mathbb{Z},$$

and, at strictly positive bridge variances, define

$$Y^*_{m,n;\boldsymbol{\epsilon}} := p_{\tau_n}(\cdot - y) \triangleleft \mathcal{Z}^*_{n-1} \bullet_{\epsilon_{n-1}} \cdots \bullet_{\epsilon_{m+1}} \mathcal{Z}^*_m \triangleright 1. \tag{51}$$

For $S \subset V$, let

$$W_k^S = \begin{cases} Q_{k,y}, & k \in S, \\ \mathcal{Z}_k, & k \notin S, \end{cases}$$

and set

$$T_{S,\boldsymbol{\epsilon}} := p_{\tau_n}(\cdot - y) \triangleleft W^S_{n-1} \bullet_{\epsilon_{n-1}} \cdots \bullet_{\epsilon_{m+1}} W^S_m \triangleright 1. \tag{52}$$

Finite multilinearity gives

$$Y^*_{m,n;\boldsymbol{\epsilon}} = \sum_{S \subset V} \left(\prod_{k \in S} c_k \right) T_{S,\boldsymbol{\epsilon}}. \tag{53}$$

Each $T_{S,\boldsymbol{\epsilon}}$ is a positive word covered by Step 2.

For later use we identify its limit explicitly. If $J = [a, d]$ is a connected component of $V \setminus S$, define

$$Y_J(y) := Y_{a,d+1}(y) = p_{\tau_{d+1},y} \triangleleft Z^{\theta}_{\tau_{d+1},\tau_a} \triangleright 1.$$

At positive bridge variances the corresponding factor is

$$Y_{J,\boldsymbol{\epsilon}}(y) := q_{J,\boldsymbol{\epsilon}} \triangleleft \mathcal{Z}_d \bullet_{\epsilon_d} \cdots \bullet_{\epsilon_{a+1}} \mathcal{Z}_a \triangleright 1,$$

where

$$q_{J,\boldsymbol{\epsilon}} = \begin{cases} p_{\tau_n}(\cdot - y), & d = n-1, \\ p_{\tau_{d+1}+\epsilon_{d+1}}(\cdot - y), & d < n-1. \end{cases}$$

Repeatedly cutting at the Q-positions gives

$$T_{S,\boldsymbol{\epsilon}} = \prod_{J \in \mathrm{comp}(V \setminus S)} Y_{J,\boldsymbol{\epsilon}}, \tag{54}$$

with the empty product equal to 1. Step 1 gives $Y_{J,\boldsymbol{\epsilon}} \to Y_J$ in L^1, and therefore Step 2 gives

$$T_{S,\boldsymbol{\epsilon}} \longrightarrow \prod_{J \in \mathrm{comp}(V \setminus S)} Y_J \qquad \text{in } L^1.$$

Since $0 \le c_k \le 1/2$, multiplication by $\prod_{k \in S} c_k$ preserves L^1 convergence even though the c_k are random. There are only finitely many subsets $S \subset V$, so (53) has an L^1 limit. Define

$$Y^*_{m,n} := L^1 - \lim_{\boldsymbol{\epsilon} \to 0} Y^*_{m,n;\boldsymbol{\epsilon}}.$$

Then

$$\boxed{Y^*_{m,n} = \sum_{S \subset V} \left(\prod_{k \in S} c_k \right) \prod_{J \in \mathrm{comp}(V \setminus S)} Y_J.} \tag{55}$$

This proves, in particular, the asserted existence in L^1 of the tested Gaussian heat-bridge compositions of the $\mathcal{Z}^*_k$.

The factors M_J associated with distinct components use disjoint SHF slabs, hence are independent and have mean one. Thus

$$\mathbb{E} Y^*_{m,n} \le \sum_{S \subset V} 2^{-|S|} = (3/2)^{n-m} < \infty.$$

For $S = \varnothing$, the only component is $V = [m, n-1]$, and the corresponding term is exactly

$$Y_V = Y_{m,n}.$$

Every other term in (55) is nonnegative. Therefore

$$0 < Y_{m,n} \le Y^*_{m,n} < \infty. \tag{56}$$

Step 4: signed words and the interval numerators. We now turn to the signed kernels

$$D_{k,y} = \mathcal{Z}_k - R_{k,y}.$$

Consider any finite regularized tested word whose factors belong to

$$\{\mathcal{Z}_k, R_{k,y}, Q_{k,y}, D_{k,y}\}.$$

If the word contains r factors D_k, expand each $D_k = \mathcal{Z}_k - R_k$. By finite multilinearity, the word becomes a finite signed sum of at most 2^r positive words with factors in $\{\mathcal{Z}_k, R_{k,y}, Q_{k,y}\}$. Every positive word has an L^1 bridge limit by Step 2. Hence the original signed word also has a well-defined L^1 bridge limit. At fixed positive bridge variances it is absolutely integrable, since its absolute value is bounded by the sum of the finitely many positive words in this expansion.

In particular, for

$$J = [j, j+h-1], \qquad h \geq 2,$$

the regularized signed chain

$$A_{J,\boldsymbol{\epsilon}}(y) := p_{j+h,y} \triangleleft D_{j+h-1,y} \bullet_{\epsilon_{j+h-1}} \cdots D_{j+1,y} \bullet_{\epsilon_{j+1}} \mathcal{Z}_j \triangleright 1$$

has an L^1 limit, and this limit is precisely the A_J of (13). There are $h-1$ factors D, so the expansion has at most 2^{h-1} positive unit-mean terms. Consequently

$$\|A_J\|_1 \leq 2^{h-1} = 2^{|J|-1}. \tag{57}$$

This is only a well-definedness bound; no cancellation is used. The same finite reset expansion and Gaussian bridge limit is the one used in the Gu–Tsai signed-chain object, so no new signed quantity is being introduced.

Step 5: the modified reset error and the interval numerators. Since

$$Q_{k,y} \triangleright 1 = \mathrm{d}x,$$

we have, as equalities of locally finite measures,

$$\mathcal{Z}_k^* \triangleright 1 = \mu_k + c_k \,\mathrm{d}x.$$

Therefore the reset kernel associated with $\mathcal{Z}_k^*$ is

$$R_{k,y}^* := (\mathcal{Z}_k^* \triangleright 1) \otimes p_{k,y} = R_{k,y} + c_k Q_{k,y}.$$

It follows immediately that

$$\boxed{D_{k,y}^* := \mathcal{Z}_k^* - R_{k,y}^* = \mathcal{Z}_k - R_{k,y} = D_{k,y}.} \tag{58}$$

Moreover,

$$D_{k,y} \triangleright 1 = 0,$$

and

$$\begin{aligned} p_{k+1,y} \triangleleft \mathcal{Z}_k^* \triangleright 1 &= p_{k+1,y} \triangleleft \mathcal{Z}_k \triangleright 1 + c_k p_{k+1,y} \triangleleft Q_{k,y} \triangleright 1 \\ &= G_k + c_k = g_k^*, \end{aligned}$$

because $p_{k+1,y} \triangleleft Q_{k,y} \triangleright 1 = 1$. This proves the one-block assertion and (40).

It remains to check that the interval numerator is unchanged when the terminal factor $\mathcal{Z}_j$ is replaced by $\mathcal{Z}_j^*$. At positive bridge variances, for $h \geq 2$ and for every Gaussian incoming profile f arising above,

$$\begin{aligned} f \triangleleft D_{j+h-1} \bullet_{\epsilon_{j+h-1}} \cdots D_{j+1} \bullet_{\epsilon_{j+1}} \mathcal{Z}_j^* \triangleright 1 \\ = f \triangleleft D_{j+h-1} \bullet_{\epsilon_{j+h-1}} \cdots D_{j+1} \bullet_{\epsilon_{j+1}} \mathcal{Z}_j \triangleright 1. \end{aligned} \tag{59}$$

Indeed, the difference is

$$c_j \, f \triangleleft D_{j+h-1} \bullet_{\epsilon_{j+h-1}} \cdots D_{j+1} \bullet_{\epsilon_{j+1}} Q_{j,y} \triangleright 1.$$

Since $Q_{j,y} \triangleright 1 = \mathrm{d}x$ and $p_{\epsilon_{j+1}} * 1 = 1$, the factor immediately to its left is $D_{j+1} \triangleright 1 = 0$, so the difference vanishes identically. Step 4 justifies this calculation by absolute integrability. Passing to the bridge limit shows that every modified interval numerator of length at least two is exactly the original A_J.

Step 6: the exact modified gas identity. At positive bridge variances write

$$\mathcal{Z}_k^* = R_{k,y}^* + D_{k,y}.$$

Every signed word in the resulting finite expansion has an L^1 bridge limit: expand

$$R_k^* = R_k + c_k Q_k, \qquad D_k = \mathcal{Z}_k - R_k,$$

and invoke Steps 2 and 4.

A word whose rightmost factor is D_m vanishes because $D_m \triangleright 1 = 0$. Every nonvanishing word has a unique decomposition, read from larger indices toward smaller indices, into consecutive blocks of the form

$$D_d \bullet \cdots \bullet D_{a+1} \bullet R_a^*.$$

These blocks form a tiling of $V = [m, n-1]$ by consecutive integer intervals. For a singleton block $J = \{a\}$, the block weight is the one-block mass g_a^*. If $J = [a, d]$ has length at least two, the rank-one factorization of R_a^* closes the block with the right mass of $\mathcal{Z}_a^*$; by (59), its bridge-limit weight is exactly A_J. Therefore

$$Y_{m,n}^* = \sum_{\mathcal{T} \text{ a tiling of } V} \prod_{J \in \mathcal{T}} B_J, \qquad B_J = \begin{cases} g_k^*, & J = \{k\}, \\ A_J, & |J| \geq 2. \end{cases} \tag{60}$$

Deleting the singleton blocks gives a bijection between such tilings and pairwise disjoint families of intervals of length at least two contained in V; adjacent nonsingleton intervals are allowed. Hence

$$Y_{m,n}^* = \prod_{k \in V} g_k^* + \sum_{I \in \mathcal{I}([m,n])} \left(\prod_{J \in I} A_J \right) \prod_{k \in V \setminus \bigcup_{J \in I} J} g_k^*.$$

Since every $g_k^* \geq 1/2$, division by $\prod_{k=m}^{n-1} g_k^*$ is legitimate and yields

$$\frac{Y_{m,n}^*}{\prod_{k=m}^{n-1} g_k^*} = 1 + \sum_{I \in \mathcal{I}([m,n])} \prod_{J \in I} \frac{A_J}{\prod_{k \in J} g_k^*} = 1 + \sum_{I \in \mathcal{I}([m,n])} \prod_{J \in I} a_J^*.$$

Together with (56), this proves (42).

Finally, all bridge convergences above are L^1 for each fixed finite integer interval. For fixed y there are only countably many such intervals, so the limiting identities may be placed on one common probability-one event. Jointly measurable spatial versions and Fubini give the corresponding almost-everywhere-in-y statements needed later. □

The next theorem packages the deterministic gas bounds, the tamed activity sums, and the clipping penalty into the one-sided estimate needed to control positive logarithmic spikes uniformly in space.

Theorem 3.4 (Whole-space positive-part decoupling)**.** *For b sufficiently small,*

$$\boxed{\sup_y \mathbb{E} \sup_{n>m\geq M} (\Lambda_{m,n}(y))_+ \leq CM^{-1/2}.} \tag{61}$$

Consequently, for every compact K,

$$\mathbb{E}\int_K \sup_{n>m\geq M} (\Lambda_{m,n}(y))_+ \,\mathrm{d}y \leq C|K|M^{-1/2}.$$

Thus (61) *is proved for the actual composed SHF on the whole spatial set, without deleting any exceptional set.*

Proof. Let $\mathcal{P}^*_{m,n}$ be the gas in (42), that is,

$$\mathcal{P}^*_{m,n} = \frac{Y^*_{m,n}}{\prod_{k=m}^{n-1} g^*_k} = 1 + \sum_{I\in\mathcal{I}([m,n])} \prod_{J\in I} a^*_J.$$

By Lemma 3.3, $\mathcal{P}^*_{m,n} > 0$. Therefore Lemma 2.5 applies and gives

$$\log \mathcal{P}^*_{m,n} \leq S^*_{m,n} + C_\star H^*_M.$$

Since $Y \leq Y^*$,

$$\begin{aligned}
\Lambda_{m,n} &= \log Y_{m,n} - \sum_{k=m}^{n-1} \log G_k \\
&\leq \log Y^*_{m,n} - \sum_{k=m}^{n-1} \log G_k \\
&= \log \mathcal{P}^*_{m,n} + \sum_{k=m}^{n-1} \log(g^*_k/G_k) \\
&\leq S^*_{m,n} + C_\star H^*_M + \left|\sum_{k=m}^{n-1} \log(g^*_k/G_k)\right|.
\end{aligned}$$

Take the positive part and the supremum over $n > m \geq M$. By Cauchy–Schwarz and Lemma 3.2,

$$\mathbb{E}\sup_{n>m\geq M} |S^*_{m,n}| \leq CM^{-1/2}, \qquad \mathbb{E}H^*_M \leq CM^{-1}.$$

Lemma 3.1 makes $\mathbb{E}\left|\sum_{k=m}^{n-1} \log(g^*_k/G_k)\right|$ smaller than any prescribed power of M^{-1}. This proves (61); Tonelli gives the spatial statement. □

4 Gaussian convexity, second moments, and the negative-part control

4.1 Gaussian convexity and deterministic-profile SHF masses

This subsection supplies the lower-tail information missing from the signed-gas analysis. The signed expansion is very effective for positive deviations of $\Lambda_{m,n}$, but large negative logarithmic spikes are more naturally controlled through a convexity property of positive Gaussian partition functions.

The external convexity input is due to Chen [4, Theorem 1]; see also Theorem A.5 in Appendix A. In the form used here, if f is convex on a finite-dimensional Gaussian space and $\mathbb{E}e^{uf} < \infty$ for every $u > 0$, then

$$\Phi(u) = \frac{1}{u} \log \mathbb{E}e^{uf}, \qquad \Phi(0) = \mathbb{E}f,$$

is convex for all real u.

The following lemma extracts the precise lower-tail information needed from Chen's Gaussian convexity theorem. It turns only a normalized second moment into inverse moments, a free-energy lower bound, and a Gaussian-type lower-tail estimate.

Lemma 4.1 (Negative moments from a second moment)**.** *Let $X > 0$, $\mathbb{E}X = 1$, and suppose that $\log X$ is a convex function of a finite-dimensional Gaussian vector, with $\mathbb{E}X^u < \infty$ for every $u > 0$. Then, for every $a > 0$,*

$$\mathbb{E}X^{-a} \le (\mathbb{E}X^2)^{a(a+1)/2}, \qquad \mathbb{E}\log X \ge -\frac{1}{2}\log(\mathbb{E}X^2). \tag{62}$$

If $\mathbb{E}X^2 > 1$, then also

$$\mathbb{P}\left(\log X < -\frac{1}{2}\log(\mathbb{E}X^2) - r\right) \le \exp\left(-\frac{r^2}{2\log(\mathbb{E}X^2)}\right), \qquad r > 0. \tag{63}$$

Proof. Apply Chen's theorem to $f = \log X$. Since $\Phi(1) = 0$ and $\Phi(2) = \frac{1}{2}\log(\mathbb{E}X^2)$, monotonicity of secant slopes gives

$$\frac{\Phi(1) - \Phi(-a)}{1 + a} \le \Phi(2) - \Phi(1) = \frac{1}{2}\log(\mathbb{E}X^2).$$

Hence $\Phi(-a) \ge -(1 + a)\log(\mathbb{E}X^2)/2$. Multiplication by the negative number $-a$ proves the inverse-moment inequality. Convexity at $0, 1, 2$ yields

$$0 = \Phi(1) \le \frac{1}{2}\Phi(0) + \frac{1}{2}\Phi(2),$$

which proves the free-energy inequality. Markov's inequality gives the right side $\exp(a^2 \log(\mathbb{E}X^2)/2 - ar)$ in (63); choose $a = r/\log(\mathbb{E}X^2)$. □

The next proposition transfers the finite-dimensional Gaussian inequality to actual SHF masses through Tsai's noise-mollified approximations. This is the step that makes Gaussian convexity available even though the limiting SHF is not itself represented as a convex functional of one Gaussian vector.

Proposition 4.2 (Transfer to SHF)**.** *Let*

$$F = f \triangleleft Z^{\theta}_{u,v} \triangleright h$$

with deterministic nonnegative profiles for which

$$\mathbb{E}F \in (0,\infty), \qquad \mathbb{E}F^2 < \infty.$$

The following assertions apply to nonnegative compact smooth profiles, and to their monotone extensions, in particular heat profiles and terminal 1*:*

$$\boxed{\mathbb{E}(F/\mathbb{E}F)^{-a} \le \left(\frac{\mathbb{E}F^2}{(\mathbb{E}F)^2}\right)^{a(a+1)/2}, \qquad \mathbb{E}\log(F/\mathbb{E}F) \ge -\frac{1}{2}\log\frac{\mathbb{E}F^2}{(\mathbb{E}F)^2}.} \tag{64}$$

Thus all negative moments of each such fixed SHF mass are finite.

Proof. We first take f, h nonnegative, compactly supported, and smooth. Let F_η be the tested mass of Tsai's noise-mollified Gaussian SHE approximation at a fixed mollification scale $\eta > 0$. These masses converge in law to F, and their first and second moments converge to those of F.

For completeness, we spell out the Feynman–Kac structure that will be used. Write ρ_η for the spatial mollifier appearing in the mollified noise, and write β_η for the corresponding deterministic coupling constant. The underlying Gaussian noise may be realized as an isonormal Gaussian process

$$W(g), \qquad g \in \mathcal{H}, \qquad \mathcal{H} = L^2([u,v]\times\mathbb{R}^2),$$

so that

$$\mathbb{E}[W(g)W(\widetilde{g})] = \langle g, \widetilde{g}\rangle_{\mathcal{H}}.$$

For a Brownian path $\omega = (\omega_s)_{u\le s\le v}$, define its mollified occupation profile

$$h_\omega(s,z) = \beta_\eta\, \rho_\eta(z-\omega_s)\mathbf{1}_{[u,v]}(s).$$

Then

$$W(h_\omega) = \beta_\eta \int_u^v \int_{\mathbb{R}^2} \rho_\eta(z-\omega_s)\, W(\,\mathrm{d}s,\,\mathrm{d}z),$$

which is exactly the stochastic integral of the mollified noise along the path. In particular $W(h_\omega)$ is centered Gaussian with variance $\|h_\omega\|_{\mathcal{H}}^2$.

The important point is that, for fixed η and fixed time interval, $\|h_\omega\|_{\mathcal{H}}$ is bounded uniformly in the path. Indeed, translation invariance of the $L^2(\mathbb{R}^2)$-norm gives

$$\begin{aligned}\|h_\omega\|_{\mathcal{H}}^2 &= \beta_\eta^2 \int_u^v \int_{\mathbb{R}^2} \left|\rho_\eta(z-\omega_s)\right|^2 \,\mathrm{d}z\,\mathrm{d}s \\ &= \beta_\eta^2 \int_u^v \|\rho_\eta\|_{L^2(\mathbb{R}^2)}^2 \,\mathrm{d}s \\ &= \beta_\eta^2 (v-u)\,\|\rho_\eta\|_{L^2(\mathbb{R}^2)}^2 .\end{aligned}$$

Hence

$$\sup_\omega \|h_\omega\|_{\mathcal{H}} \le C_\eta := |\beta_\eta|\sqrt{v-u}\;\|\rho_\eta\|_{L^2(\mathbb{R}^2)} < \infty. \tag{65}$$

No uniformity as $\eta \downarrow 0$ is needed here.

The Feynman–Kac formula can therefore be written in the form

$$F_\eta = \int \nu(\,\mathrm{d}\omega) \exp\left\{W(h_\omega) - \frac{1}{2}\|h_\omega\|_{\mathcal{H}}^2\right\}, \tag{66}$$

where ν is a finite deterministic positive Brownian path measure. More explicitly, if the path is represented by its endpoints x, y and a Brownian bridge from x to y over the time interval $[u, v]$, then ν has the form

$$\nu(\,\mathrm{d}\omega) = f(x)\, p_{v-u}(y-x)\, h(y)\; \mathrm{d}x\; \mathrm{d}y\, \mathbb{P}_{u,v}^{x\to y}(\,\mathrm{d}\omega),$$

up to the notational convention used to encode the path and its endpoints. Consequently,

$$\nu(\Omega) = \iint_{\mathbb{R}^2\times\mathbb{R}^2} f(x) p_{v-u}(y-x) h(y)\; \mathrm{d}x\; \mathrm{d}y. \tag{67}$$

On the other hand, the first moment of the SHF kernel is the deterministic heat kernel,

$$\mathbb{E} Z_{u,v}^{\theta}(\,\mathrm{d}x,\, \mathrm{d}y) = p_{v-u}(y-x)\; \mathrm{d}x\; \mathrm{d}y,$$

and therefore

$$\mathbb{E}F = \iint_{\mathbb{R}^2\times\mathbb{R}^2} f(x) p_{v-u}(y-x) h(y)\; \mathrm{d}x\; \mathrm{d}y = \nu(\Omega). \tag{68}$$

The same identity can also be read directly from (66): since $W(h_\omega) \sim N(0, \|h_\omega\|_{\mathcal{H}}^2)$,

$$\mathbb{E}\exp\left\{W(h_\omega) - \frac{1}{2}\,\|h_\omega\|_{\mathcal{H}}^2\right\} = 1,$$

and hence $\mathbb{E}F_\eta = \nu(\Omega) = \mathbb{E}F$.

Now let Π_d be increasing finite-dimensional orthogonal projections in $\mathcal{H}$, and choose an orthonormal basis $e_1, \ldots, e_d$ of $\operatorname{Ran}\Pi_d$. Set

$$\xi_j = W(e_j), \qquad \xi = (\xi_1, \ldots, \xi_d) \sim N(0, I_d).$$

Define

$$F_{\eta,d} = \mathbb{E}[F_\eta \mid \xi_1, \ldots, \xi_d].$$

Using the orthogonal decomposition

$$h_\omega = \Pi_d h_\omega + (I - \Pi_d) h_\omega$$

and independence of the corresponding Gaussian components, together with

$$\mathbb{E}\exp\left\{W((I-\Pi_d)h_\omega) - \frac{1}{2}\,\|(I-\Pi_d)h_\omega\|^2\right\} = 1,$$

we obtain

$$F_{\eta,d} = \int \nu(\,\mathrm{d}\omega) \exp\left\{\langle \Pi_d h_\omega, \xi\rangle - \frac{1}{2}\,\|\Pi_d h_\omega\|^2\right\}. \tag{69}$$

In particular, by the tower property and $\mathbb{E}F_\eta = \mathbb{E}F$,

$$\mathbb{E}F_{\eta,d} = \mathbb{E}\,\mathbb{E}[F_\eta \mid \xi_1, \ldots, \xi_d] = \mathbb{E}F_\eta = \mathbb{E}F. \tag{70}$$

We next verify the hypotheses needed for the finite-dimensional Gaussian convexity lemma. For

$$L_\omega(\xi) = \langle \Pi_d h_\omega, \xi\rangle - \frac{1}{2}\,\|\Pi_d h_\omega\|^2,$$

the map $\xi \mapsto L_\omega(\xi)$ is affine. Hence, for $\xi, \zeta \in \mathbb{R}^d$ and $\lambda \in [0,1]$,

$$\begin{aligned} F_{\eta,d}(\lambda\xi + (1-\lambda)\zeta) &= \int \left(e^{L_\omega(\xi)}\right)^\lambda \left(e^{L_\omega(\zeta)}\right)^{1-\lambda} \nu(\,\mathrm{d}\omega) \\ &\le F_{\eta,d}(\xi)^\lambda F_{\eta,d}(\zeta)^{1-\lambda} \end{aligned}$$

by Hölder's inequality. Taking logarithms shows that $\log F_{\eta,d}$ is convex.

The uniform bound (65) also gives the required integrability. Since orthogonal projection does not increase norm,

$$\|\Pi_d h_\omega\| \le C_\eta.$$

For any $\xi, \zeta \in \mathbb{R}^d$,

$$\left|\langle \Pi_d h_\omega, \xi - \zeta\rangle\right| \le C_\eta \left\|\xi - \zeta\right\|,$$

uniformly in ω. Comparing the two integrands in (69) therefore yields

$$e^{-C_\eta\|\xi-\zeta\|} F_{\eta,d}(\zeta) \le F_{\eta,d}(\xi) \le e^{C_\eta\|\xi-\zeta\|} F_{\eta,d}(\zeta),$$

and hence

$$\left|\log F_{\eta,d}(\xi) - \log F_{\eta,d}(\zeta)\right| \le C_\eta \left\|\xi - \zeta\right\|. \tag{71}$$

Thus $\log F_{\eta,d}$ is globally Lipschitz. Taking $\zeta = 0$,

$$|\log F_{\eta,d}(\xi)| \le |\log F_{\eta,d}(0)| + C_\eta \left\|\xi\right\|.$$

Moreover,

$$e^{-C_\eta^2/2}\nu(\Omega) \le F_{\eta,d}(0) = \int e^{-\|\Pi_d h_\omega\|^2/2}\, \nu(\,\mathrm{d}\omega) \le \nu(\Omega),$$

so $F_{\eta,d}(0) \in (0,\infty)$. Since a finite-dimensional Gaussian vector satisfies $\mathbb{E}\left\|\xi\right\| < \infty$, we conclude that

$$\mathbb{E}|\log F_{\eta,d}| < \infty.$$

Likewise, for every $p > 0$, (71) gives

$$F_{\eta,d}(\xi)^p \le F_{\eta,d}(0)^p e^{pC_\eta\|\xi\|}.$$

A finite-dimensional Gaussian vector has finite exponential moments of its norm:

$$\mathbb{E}e^{c\|\xi\|} < \infty \qquad \text{for every } c > 0.$$

Therefore

$$\mathbb{E}F_{\eta,d}^p < \infty \qquad \text{for every } p > 0. \tag{72}$$

We may now apply the finite-dimensional Gaussian lemma to

$$X_{\eta,d} := \frac{F_{\eta,d}}{\mathbb{E}F}.$$

By (70), $\mathbb{E}X_{\eta,d} = 1$; by the argument above, $\log X_{\eta,d}$ is convex and all positive moments are finite. Hence, for every $a > 0$,

$$\mathbb{E}X_{\eta,d}^{-a} \le \left(\mathbb{E}X_{\eta,d}^2\right)^{a(a+1)/2}. \tag{73}$$

For fixed η, the variables $F_{\eta,d}$ form the martingale

$$F_{\eta,d} = \mathbb{E}[F_\eta \mid \xi_1, \dots, \xi_d],$$

and $F_{\eta,d} \to F_\eta$ in L^2 by martingale convergence. For each $a > 0$, the function

$$x \longmapsto \begin{cases} x^{-a}, & x > 0, \\ +\infty, & x = 0, \end{cases}$$

is nonnegative and lower semicontinuous on $[0, \infty)$. Therefore, if $X_n \Rightarrow X$, Portmanteau gives

$$\mathbb{E}X^{-a} \le \liminf_{n\to\infty} \mathbb{E}X_n^{-a}.$$

Applying this to $d \to \infty$, while using $\mathbb{E}F_{\eta,d}^2 \to \mathbb{E}F_\eta^2$, yields

$$\mathbb{E}(F_\eta/\mathbb{E}F)^{-a} \le \left(\frac{\mathbb{E}F_\eta^2}{(\mathbb{E}F)^2}\right)^{a(a+1)/2}.$$

We then let $\eta \downarrow 0$. The tested mollified masses converge in law to F, while their first and second moments converge to those of F. Applying the same lower-semicontinuity argument once more gives

$$\mathbb{E}(F/\mathbb{E}F)^{-a} \le \left(\frac{\mathbb{E}F^2}{(\mathbb{E}F)^2}\right)^{a(a+1)/2}.$$

Since the left-hand side would be infinite if $F = 0$ with positive probability, this also proves $F > 0$ almost surely.

The second-moment assumption controls the positive part of $\log F$, while the inverse first moment controls the negative part, so $\log(F/\mathbb{E}F)$ is integrable. For any $a > 0$, Jensen's inequality gives

$$\mathbb{E}\log(F/\mathbb{E}F) = -\frac{1}{a}\mathbb{E}\log(F/\mathbb{E}F)^{-a} \ge -\frac{1}{a}\log\mathbb{E}(F/\mathbb{E}F)^{-a}.$$

Using the inverse-moment estimate,

$$\mathbb{E}\log(F/\mathbb{E}F) \ge -\frac{a+1}{2}\log\frac{\mathbb{E}F^2}{(\mathbb{E}F)^2}.$$

Letting $a \downarrow 0$ proves the second inequality in (64).

Finally, approximate a heat profile and terminal 1 from below by nonnegative compact smooth profiles. The corresponding masses increase almost surely, their means increase, and their second moments increase to those of the target mass. Fatou's lemma passes the inverse-moment estimate once more. No approximation estimate uniform in auxiliary scale indices is required: the limit is taken for each fixed mass, and the resulting inequality contains only the exact limiting second moment. □

4.2 Exact second moments and their uniform asymptotics

The second-moment input used in this section consists of two standard facts from the SHF moment theory. First, the SHF moment identity of Tsai [5, Definition 1.2(d), equation (1.2)], specialized to $n = 2$, identifies the second moment of a tested SHF mass with the two-particle delta-Bose semigroup. Second, we use the explicit two-replica kernel written in Gu–Tsai [6, (2.4)–(2.6)]. We do not rederive the delta-Bose diagram expansion here. The only calculation needed below is the elementary integration of that quoted kernel against our particular heat-kernel starting profile and terminal test 1.

There is one minor technical point: Tsai's moment identity is stated for L^2 terminal test functions, whereas we use the terminal profile 1. Whenever this identity is applied below, we first replace 1 by nonnegative compactly supported cutoffs $\chi_R \uparrow 1$, apply the moment identity to χ_R, and then pass to the limit by positivity and monotone convergence. We shall use this extension without further comment.

The fourth-moment estimate (19) has a different origin: it is the $r = 4$ case of the one-block high-moment estimate quoted in Lemma 2.3. It is used later to justify the second-order Taylor expansion of $\log G_k$ with an $O(k^{-3/2})$ remainder.

Define

$$\mathsf{j}^\theta(t) = \int_0^\infty \frac{t^{a-1}e^{\theta a}}{\Gamma(a)}\,\mathrm{d}a, \qquad J_\theta(t) = \int_0^t \mathsf{j}^\theta(r)\,\mathrm{d}r = \int_0^\infty \frac{e^{-A(t)a}}{\Gamma(1+a)}\,\mathrm{d}a, \qquad A(t) = \log t^{-1} - \theta.$$

These are the conventions of [6, (2.4)–(2.6)].

Before evaluating the SHF second moments, we record the elementary asymptotic of the interaction primitive J_θ. It is used repeatedly to turn the exact two-replica formulas into uniform scale estimates.

Lemma 4.3. *As $A(t) \to \infty$,*

$$J_\theta(t) = \frac{1}{A(t)} + O(A(t)^{-2}). \tag{74}$$

Moreover J_θ is increasing and finite at every fixed positive time.

Proof. The reciprocal gamma function is bounded on $[1,\infty)$ and $\Gamma(1+a)^{-1} = 1 + O(a)$ on $[0,1]$. Subtract $\int_0^\infty e^{-Aa}\,\mathrm{d}a = 1/A$, estimate the difference on $[0,1]$ by $C\int_0^\infty ae^{-Aa}\,\mathrm{d}a$, and estimate the remaining tail exponentially by

$$\left|\int_1^\infty e^{-A(t)a}\left(\frac{1}{\Gamma(1+a)} - 1\right)\,\mathrm{d}a\right| \le C\int_1^\infty e^{-A(t)a}\,\mathrm{d}a = \frac{C}{A}e^{-A}$$

Monotonicity follows from the definition as an integral of $\mathsf{j}^\theta > 0$; finiteness at fixed time follows from the growth of $\Gamma(1+a)$. □

This lemma specializes the external two-replica moment formula to a heat-kernel starting profile and terminal test 1. Its main purpose is to identify the exact second moment of $Y_{m,n}$ and to bound it uniformly even when the ratio n/m is unbounded.

Lemma 4.4 (Heat-start mass)**.** *For $a, t > 0$, we have*

$$\mathbb{E}|p_a(\cdot - y) \triangleleft Z^\theta_{0,t} \triangleright 1|^2 = 1 + \int_0^t \frac{J_\theta(t-u)}{a+u}\,\mathrm{d}u. \tag{75}$$

In particular,

$$\mathbb{E}Y_{m,n}(y)^2 = 1 + \int_{\tau_n}^{\tau_m} \frac{J_\theta(\tau_m - u)}{u}\,\mathrm{d}u, \tag{76}$$

$$\log \mathbb{E}Y_{m,n}(y)^2 \le \log(n/m) + C/m, \qquad n > m. \tag{77}$$

Proof. By Tsai's second-moment identity [5, Definition 1.2(d), equation (1.2)], together with the explicit two-replica kernel [6, (2.4)–(2.6)], the second moment consists of the free two-replica contribution, which is 1, plus the interaction term

$$4\pi \int_{0<u<v<t} \mathsf{j}^\theta(v-u)\left(\int_{\mathbb{R}^2} p_{a+u}(x-y)^2\,\mathrm{d}x\right)\,\mathrm{d}u\,\mathrm{d}v. \tag{78}$$

Here the identity is first applied with terminal cutoffs $\chi_R \uparrow 1$ as explained above. In the quoted kernel, integration against the terminal profile removes the two terminal heat kernels, while the heat-semigroup identity

$$p_a * p_u = p_{a+u}$$

turns the two starting profiles into the square $p_{a+u}(\cdot - y)^2$. Letting $R \to \infty$ gives (78) by monotone convergence. For more details, see Proposition A.4.

The remaining spatial integral is elementary:

$$\int_{\mathbb{R}^2} p_{a+u}(x-y)^2 \,\mathrm{d}x = p_{2(a+u)}(0) = \frac{1}{4\pi(a+u)}.$$

Thus the factor 4π in the quoted two-replica kernel cancels, and

$$\mathbb{E}|p_a(\cdot - y) \triangleleft Z^\theta_{0,t} \triangleright 1|^2 = 1 + \int_{0<u<v<t} \frac{\mathsf{j}^\theta(v-u)}{a+u} \,\mathrm{d}u \,\mathrm{d}v.$$

For fixed u,

$$\int_u^t \mathsf{j}^\theta(v-u) \,\mathrm{d}v = \int_0^{t-u} \mathsf{j}^\theta(r) \,\mathrm{d}r = J_\theta(t-u),$$

which proves (75).

Now use time stationarity and set

$$a = \tau_n, \qquad t = \tau_m - \tau_n.$$

Then

$$\mathbb{E}Y_{m,n}(y)^2 = 1 + \int_0^{\tau_m - \tau_n} \frac{J_\theta(\tau_m - \tau_n - u)}{\tau_n + u} \,\mathrm{d}u.$$

With the change of variables $r = \tau_n + u$, this becomes

$$\mathbb{E}Y_{m,n}(y)^2 = 1 + \int_{\tau_n}^{\tau_m} \frac{J_\theta(\tau_m - r)}{r} \,\mathrm{d}r,$$

which is (76).

Finally, monotonicity of J_θ gives

$$\mathbb{E}Y_{m,n}(y)^2 \le 1 + J_\theta(\tau_m) \int_{\tau_n}^{\tau_m} \frac{\mathrm{d}u}{u} = 1 + 2L(n-m)J_\theta(\tau_m).$$

Since $\tau_m = b^{2m}$ and $L = \log b^{-1}$, Lemma 4.3 yields

$$2LJ_\theta(\tau_m) = \frac{1}{m} + O(m^{-2}).$$

Hence

$$\mathbb{E}Y_{m,n}(y)^2 \le \frac{n}{m} + C\frac{n-m}{m^2} \le \frac{n}{m}(1 + C/m).$$

Taking logarithms proves (77), uniformly in the unbounded ratio n/m. □

The next estimate supplies uniform integrability for the logarithms of the one-block masses. It is used both to control clipping events and to justify the Taylor expansion of $\log G_k$.

Lemma 4.5 (Uniform one-block logarithmic moments)**.** *For every fixed finite $p \geq 1$,*

$$\|\log G_k(y)\|_p \leq C_p k^{-1/2} \tag{79}$$

uniformly in y.

Proof. Fix $y \in \mathbb{R}^2$ and write

$$R_k := \mathbb{E}G_k(y)^2 = 1 + v_k.$$

Since $\mathbb{E}G_k(y) = 1$, we have

$$v_k = \mathrm{Var}(G_k(y)) = \mathbb{E}(G_k(y) - 1)^2.$$

We first show that

$$v_k \leq \frac{C}{k}. \tag{80}$$

By the definition of G_k,

$$G_k(y) = p_{\tau_{k+1}}(\cdot - y) \triangleleft Z^{\theta}_{\tau_{k+1},\tau_k} \triangleright 1,$$

so $G_k = Y_{k,k+1}$. Hence (76) gives

$$v_k = \int_{\tau_{k+1}}^{\tau_k} \frac{J_\theta(\tau_k - u)}{u}\,\mathrm{d}u.$$

Because J_θ is increasing and $0 \leq \tau_k - u \leq \tau_k$ on the interval of integration,

$$v_k \leq J_\theta(\tau_k) \int_{\tau_{k+1}}^{\tau_k} \frac{\mathrm{d}u}{u} = J_\theta(\tau_k) \log \frac{\tau_k}{\tau_{k+1}}.$$

Recalling that

$$\tau_k = b^{2k}, \qquad L = \log b^{-1},$$

we have

$$\log \frac{\tau_k}{\tau_{k+1}} = 2L.$$

Therefore

$$v_k \leq 2L\, J_\theta(\tau_k).$$

Now

$$A(\tau_k) = \log \tau_k^{-1} - \theta = 2Lk - \theta,$$

and Lemma 4.3 yields

$$J_\theta(\tau_k) = \frac{1}{2Lk - \theta} + O(k^{-2}) = O(k^{-1}).$$

This proves (80). In particular,

$$\log R_k = \log(1 + v_k) \leq v_k \leq \frac{C}{k}. \tag{81}$$

We next control the negative tail of $\log G_k$. Since G_k is a deterministic-profile SHF mass with heat starting profile and terminal profile 1, Proposition 4.2 applies. Because $\mathbb{E}G_k = 1$, for every $a > 0$ it gives

$$\mathbb{E}G_k^{-a} \leq R_k^{a(a+1)/2}. \tag{82}$$

We now repeat the same Markov optimization used in the proof of (63). For $r > 0$ and $a > 0$,

$$\left\{\log G_k < -\frac{1}{2}\log R_k - r\right\} = \left\{G_k^{-a} > R_k^{a/2} e^{ar}\right\}.$$

Hence Markov's inequality and (82) give

$$\begin{aligned}\mathbb{P}\left(\log G_k < -\frac{1}{2}\log R_k - r\right) &\le R_k^{-a/2} e^{-ar}\mathbb{E}G_k^{-a} \\ &\le \exp\left\{\frac{a^2}{2}\log R_k - ar\right\}.\end{aligned}$$

If $R_k > 1$, choose

$$a = \frac{r}{\log R_k}.$$

Then

$$\mathbb{P}\left(\log G_k < -\frac{1}{2}\log R_k - r\right) \le \exp\left(-\frac{r^2}{2\log R_k}\right), \qquad r > 0. \tag{83}$$

If $R_k = 1$, then $v_k = 0$, hence $G_k = 1$ almost surely and the desired bound is trivial.

We now convert (83) into an L^p estimate for the negative part. Put

$$X_k := (\log G_k)_- = \max\{-\log G_k, 0\}.$$

For every $s \ge \frac{1}{2}\log R_k$,

$$\{X_k > s\} = \{\log G_k < -s\} \subset \left\{\log G_k < -\frac{1}{2}\log R_k - \left(s - \frac{1}{2}\log R_k\right)\right\}.$$

Therefore

$$\mathbb{P}(X_k > s) \le \exp\left(-\frac{\left(s - \frac{1}{2}\log R_k\right)^2}{2\log R_k}\right), \qquad s \ge \frac{1}{2}\log R_k.$$

Using the layer-cake formula,

$$\mathbb{E}X_k^p = p\int_0^\infty s^{p-1}\mathbb{P}(X_k > s)\,\mathrm{d}s,$$

we split at $s = \frac{1}{2}\log R_k$. The contribution of the first part is at most

$$\left(\frac{1}{2}\log R_k\right)^p.$$

For the second part, set

$$s = \frac{1}{2}\log R_k + \sqrt{\log R_k}\, z.$$

Then

$$\int_{\frac{1}{2}\log R_k}^\infty s^{p-1}\mathbb{P}(X_k > s)\,\mathrm{d}s \le \sqrt{\log R_k}\int_0^\infty \left(\frac{1}{2}\log R_k + \sqrt{\log R_k}\, z\right)^{p-1} e^{-z^2/2}\,\mathrm{d}z.$$

Since p is fixed and $\log R_k$ is uniformly bounded for $k \ge 1$, the Gaussian moments are finite and the last display is bounded by

$$C_p(\log R_k)^{p/2}.$$

Consequently,

$$\|(\log G_k)_-\|_p \le C_p(\log R_k)^{1/2} \le C_p k^{-1/2},$$

where the last inequality follows from (81).

It remains to control the positive part. For every $x > 0$,

$$\log_+ x \le (x-1)_+ \le |x-1|.$$

Choose an even integer $q \ge p$. By (17),

$$\mathbb{E}|G_k(y) - 1|^q \le C_q k^{-q/2}.$$

Since $q \ge p$,

$$\|(\log G_k)_+\|_p \le \|G_k - 1\|_p \le \|G_k - 1\|_q \le C_p k^{-1/2}.$$

Combining the negative- and positive-part estimates,

$$\|\log G_k(y)\|_p \le \|(\log G_k(y))_-\|_p + \|(\log G_k(y))_+\|_p \le C_p k^{-1/2}.$$

All bounds are uniform in y, because the estimates entering (76), Proposition 4.2, and (17) are uniform in the spatial center. This proves (79). □

The following drift computation identifies the deterministic centering accumulated across scales. In particular, it is the source of the $-\frac{1}{2}\log(n/m)$ term that later becomes the centering c_ε.

Lemma 4.6 (Precise one-block drift)**.** *For fixed b,*

$$\mathbb{E}(G_k - 1)^2 = \frac{1}{k} + O(k^{-2}), \qquad \mathbb{E}\log G_k = -\frac{1}{2k} + O(k^{-3/2}). \tag{84}$$

Consequently,

$$\sum_{k=m}^{n-1} \mathbb{E}\log G_k = -\frac{1}{2}\log(n/m) + O(m^{-1/2}) \tag{85}$$

uniformly over $n > m$.

Proof. By (76), with

$$A_k := A(\tau_k) = 2Lk - \theta,$$

we have

$$v_k := \mathbb{E}(G_k - 1)^2 = \int_{b^2}^1 \frac{J_\theta(\tau_k(1-v))}{v}\, \mathrm{d}v.$$

Since $\tau_k = b^{2k} = e^{-2Lk}$,

$$A(\tau_k(1-v)) = \log\frac{1}{\tau_k(1-v)} - \theta = 2Lk - \theta - \log(1-v) = A_k - \log(1-v).$$

Hence Lemma 4.3 yields

$$J_\theta(\tau_k(1-v)) = \frac{1}{A_k - \log(1-v)} + O\big((A_k - \log(1-v))^{-2}\big).$$

Therefore

$$v_k = \int_{b^2}^1 \frac{1}{v(A_k - \log(1-v))}\, \mathrm{d}v + O\bigg(\int_{b^2}^1 \frac{1}{v(A_k - \log(1-v))^2}\, \mathrm{d}v\bigg).$$

Since $-\log(1-v) \geq 0$ on $v \in (0,1]$, we have $A_k - \log(1-v) \geq A_k$, and thus

$$\int_{b^2}^1 \frac{1}{v(A_k - \log(1-v))^2}\,\mathrm{d}v \leq \frac{1}{A_k^2}\int_{b^2}^1 \frac{\mathrm{d}v}{v} = \frac{2L}{A_k^2}.$$

Thus

$$v_k = \int_{b^2}^1 \frac{1}{v(A_k - \log(1-v))}\,\mathrm{d}v + O_b(A_k^{-2}).$$

We next replace $(A_k - \log(1-v))^{-1}$ by A_k^{-1}. Since $-\log(1-v) \geq 0$,

$$\left|\frac{1}{A_k - \log(1-v)} - \frac{1}{A_k}\right| = \frac{-\log(1-v)}{A_k(A_k - \log(1-v))} \leq \frac{-\log(1-v)}{A_k^2}.$$

Moreover,

$$\int_{b^2}^1 \frac{-\log(1-v)}{v}\,\mathrm{d}v = \int_{b^2}^1 \frac{-\log(1-v)}{v}\,\mathrm{d}v < \infty,$$

because v^{-1} is bounded on $[b^2,1]$ and $-\log(1-v)$ is integrable at $v=1$. Consequently,

$$\int_{b^2}^1 \frac{1}{v(A_k - \log(1-v))}\,\mathrm{d}v = \frac{1}{A_k}\int_{b^2}^1 \frac{\mathrm{d}v}{v} + O_b(A_k^{-2}).$$

Since

$$\int_{b^2}^1 \frac{\mathrm{d}v}{v} = -\log b^2 = 2L,$$

we obtain

$$v_k = \frac{2L}{A_k} + O_b(A_k^{-2}).$$

Finally, because $A_k = 2Lk - \theta$,

$$\frac{2L}{A_k} = \frac{1}{k}\frac{1}{1-\theta/(2Lk)} = \frac{1}{k} + O_b(k^{-2}),$$

and $A_k^{-2} = O_b(k^{-2})$. Therefore

$$\boxed{v_k = \frac{1}{k} + O_b(k^{-2}).}$$

On $G_k \geq 1/2$, Taylor's theorem gives

$$\left|\log G_k - (G_k - 1) + \tfrac{1}{2}(G_k-1)^2\right| \leq C|G_k - 1|^3.$$

The expectation of the cubic remainder is $O(k^{-3/2})$: by Hölder, $\mathbb{E}|G_k-1|^3 \leq (\mathbb{E}|G_k-1|^2)^{1/2}(\mathbb{E}|G_k-1|^4)^{1/2} = O(k^{-3/2})$ using (19) and Lemma 4.5. On the exceptional event $\{G_k < 1/2\}$, we first note that

$$\{G_k < 1/2\} \subset \{|G_k - 1| > 1/2\}.$$

Hence the fourth-moment estimate (19) and Markov's inequality give

$$\mathbb{P}(G_k < 1/2) \leq \mathbb{P}(|G_k - 1| > 1/2) \leq 16\,\mathbb{E}|G_k - 1|^4 \leq Ck^{-2}.$$

For the logarithmic term, Cauchy–Schwarz together with Lemma 4.5 yields

$$\begin{aligned}\mathbb{E}\left[|\log G_k|\,\mathbf{1}_{\{G_k<1/2\}}\right] &\leq \|\log G_k\|_2\,\mathbb{P}(G_k < 1/2)^{1/2} \\ &\leq Ck^{-1/2}\cdot Ck^{-1} = O(k^{-3/2}).\end{aligned}$$

The polynomial terms are even smaller on this event. Indeed, since $0 < G_k < 1/2$ there,

$$-1 < G_k - 1 < -\frac{1}{2},$$

and therefore

$$|G_k - 1| \le 1, \qquad (G_k - 1)^2 \le 1.$$

Consequently,

$$\mathbb{E}\big[|G_k - 1|\,\mathbf{1}_{\{G_k<1/2\}}\big] + \mathbb{E}\big[(G_k - 1)^2\,\mathbf{1}_{\{G_k<1/2\}}\big] \le 2\,\mathbb{P}(G_k < 1/2) = O(k^{-2}).$$

Combining these estimates with the Taylor bound on $\{G_k \ge 1/2\}$, we obtain

$$\mathbb{E}\left|\log G_k - (G_k - 1) + \frac{1}{2}(G_k - 1)^2\right| = O(k^{-3/2}).$$

Since $\mathbb{E}G_k = 1$, we have

$$\mathbb{E}(G_k - 1) = 0.$$

Therefore

$$\begin{aligned}\mathbb{E}\log G_k &= \mathbb{E}(G_k - 1) - \frac{1}{2}\mathbb{E}(G_k - 1)^2 + O(k^{-3/2}) \\ &= -\frac{1}{2}\mathbb{E}(G_k - 1)^2 + O(k^{-3/2}).\end{aligned}$$

Using the variance asymptotic already proved,

$$\mathbb{E}(G_k - 1)^2 = \frac{1}{k} + O(k^{-2}),$$

we conclude that

$$\mathbb{E}\log G_k = -\frac{1}{2k} + O(k^{-3/2}),$$

where the $O(k^{-2})$ term has been absorbed into $O(k^{-3/2})$.

Finally,

$$\sum_{k=m}^{n-1} \frac{1}{k} = \log\frac{n}{m} + O(m^{-1}),$$

uniformly for $n > m$, while

$$\sum_{k=m}^{n-1} k^{-3/2} \le \sum_{k\ge m} k^{-3/2} = O(m^{-1/2}).$$

Hence

$$\sum_{k=m}^{n-1} \mathbb{E}\log G_k = -\frac{1}{2}\log\frac{n}{m} + O(m^{-1/2}),$$

uniformly over $n > m$. □

4.3 Control of the negative-spike

The positive part of $\Lambda_{m,n}$ was controlled in Section 3, but this alone cannot rule out rare, very negative logarithmic spikes. The free-energy lower bound from Gaussian convexity supplies the missing signed information at the level of the mean. The identity $\mathbb{E}|X| = 2\mathbb{E}X_+ - \mathbb{E}X$ then converts a one-sided positive-part estimate plus a lower bound on $\mathbb{E}X$ into a full L^1 bound. This is the mechanism that closes the obstruction.

Proof of Theorem 2.1. Proposition 4.2 and Lemma 4.4 give

$$\mathbb{E}\log Y_{m,n} \geq -\frac{1}{2}\log \mathbb{E}Y_{m,n}^2 \geq -\frac{1}{2}\log(n/m) - C/m.$$

Subtract (85). This proves (6). By Theorem 3.4, equivalently (61),

$$\mathbb{E}|\Lambda_{m,n}| = 2\mathbb{E}(\Lambda_{m,n})_+ - \mathbb{E}\Lambda_{m,n} \leq Cm^{-1/2}.$$

Translation invariance and Tonelli prove (7). □

In particular, for any bounded compactly supported φ, without a mean-zero condition,

$$\sup_{n>m\geq M} \mathbb{E}|\langle \Lambda_{m,n}, \varphi\rangle| \leq C\,\|\varphi\|_1\, M^{-1/2}. \tag{86}$$

5 From microscopic windows to the original regularized field

5.1 A centered fine-window estimate after spatial testing

The next lemma is the spatially tested version of the aligned fine-window estimate. It separates the linear fluctuation, where spatial covariance can be used, from the nonlinear logarithmic remainder, where a pointwise L^2 bound and Minkowski's inequality suffice.

Lemma 5.1. *For all real $\varphi \in C_c^\infty(\mathbb{R}^2)$,*

$$\sup_{n>m\geq M} \mathbb{E}\left|\left\langle \log Y_{m,n} + \tfrac{1}{2}\log(n/m), \varphi\right\rangle\right| \leq \frac{C}{\sqrt{M}}\big(\,\|\varphi\|_1 + \|\varphi\|_2\,\big). \tag{87}$$

No mean-zero condition is needed because of the displayed deterministic centering.

Proof. Put

$$T_k(y) = \log G_k(y) - (G_k(y) - 1).$$

We claim that, uniformly in y,

$$\|T_k(y)\|_{L^2(\Omega)} \leq \frac{C}{k}. \tag{88}$$

We decompose directly according to the events $\{G_k(y) \geq \frac{1}{2}\}$ and $\{G_k(y) < \frac{1}{2}\}$:

$$\mathbb{E}|T_k(y)|^2 = \mathbb{E}\big[|T_k(y)|^2 \mathbf{1}_{\{G_k(y)\geq 1/2\}}\big] + \mathbb{E}\big[|T_k(y)|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big].$$

For $x \geq 1/2$, Taylor's theorem applied to $\log x$ around $x = 1$ gives

$$|\log x - (x-1)| \leq C|x-1|^2.$$

Hence, on $\{G_k(y) \geq 1/2\}$,

$$|T_k(y)| = \big|\log G_k(y) - (G_k(y) - 1)\big| \leq C|G_k(y) - 1|^2.$$

Therefore

$$\mathbb{E}\big[|T_k(y)|^2 \mathbf{1}_{\{G_k(y)\geq 1/2\}}\big] \leq C\mathbb{E}|G_k(y) - 1|^4.$$

Using the one-block fourth-moment estimate

$$\mathbb{E}|G_k(y) - 1|^4 \leq Ck^{-2}, \tag{19}$$

we obtain

$$\mathbb{E}\big[|T_k(y)|^2 \mathbf{1}_{\{G_k(y)\geq 1/2\}}\big] \leq Ck^{-2}. \tag{89}$$

On the event $\{G_k(y) < 1/2\}$ we do not use a Taylor expansion, since the logarithm becomes singular as $G_k(y) \downarrow 0$. Instead, from

$$T_k(y) = \log G_k(y) - (G_k(y) - 1)$$

and $(a + b)^2 \leq 2a^2 + 2b^2$, we have

$$|T_k(y)|^2 \leq 2|\log G_k(y)|^2 + 2|G_k(y) - 1|^2.$$

Consequently

$$\mathbb{E}\big[|T_k(y)|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big] \leq 2\mathbb{E}\big[|\log G_k(y)|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big] + 2\mathbb{E}\big[|G_k(y) - 1|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big]. \tag{90}$$

We first treat the logarithmic term. By Cauchy–Schwarz,

$$\mathbb{E}\big[|\log G_k(y)|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big] \leq \big(\mathbb{E}|\log G_k(y)|^4\big)^{1/2} \mathbb{P}(G_k(y) < 1/2)^{1/2}.$$

Lemma 4.5, taken with $p = 4$, gives

$$\|\log G_k(y)\|_{L^4(\Omega)} \leq Ck^{-1/2},$$

and hence

$$\big(\mathbb{E}|\log G_k(y)|^4\big)^{1/2} = \|\log G_k(y)\|^2_{L^4(\Omega)} \leq Ck^{-1}.$$

Moreover,

$$\{G_k(y) < 1/2\} \subseteq \{|G_k(y) - 1| > 1/2\},$$

so the probability estimate in (20) yields

$$\mathbb{P}(G_k(y) < 1/2) \leq \mathbb{P}(|G_k(y) - 1| > 1/2) \leq Ck^{-2}.$$

Thus

$$\mathbb{E}\big[|\log G_k(y)|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big] \leq (Ck^{-1})(Ck^{-1}) \leq Ck^{-2}. \tag{91}$$

For the polynomial term there is an even simpler estimate. Since $0 < G_k(y) < 1/2$ on B, we have $|G_k(y) - 1| \leq 1$, and therefore

$$\mathbb{E}\big[|G_k(y) - 1|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big] \leq \mathbb{P}(G_k(y) < 1/2) \leq Ck^{-2}. \tag{92}$$

Combining (90), (91), and (92), we conclude that

$$\mathbb{E}\big[|T_k(y)|^2 \mathbf{1}_{\{G_k(y)<1/2\}}\big] \leq Ck^{-2}. \tag{93}$$

Adding (89) and (93),

$$\mathbb{E}|T_k(y)|^2 \le Ck^{-2},$$

uniformly in y. Taking square roots proves

$$\|T_k(y)\|_{L^2(\Omega)} \le \frac{C}{k},$$

as desired.

Now, Proposition (A.4) gives the two-replica formula

$$\mathrm{Cov}(G_k(y), G_k(z)) = 4\pi \int_{\tau_{k+1}}^{\tau_k} p_{2u}(y-z) J_\theta(\tau_k - u)\,\mathrm{d}u.$$

Therefore

$$\begin{aligned}\mathbb{E}|\langle G_k - 1, \varphi\rangle|^2 &= 4\pi \int_{\tau_{k+1}}^{\tau_k} J_\theta(\tau_k - u) \iint \varphi(y) p_{2u}(y-z)\varphi(z)\,\mathrm{d}y\,\mathrm{d}z\;\mathrm{d}u \\ &= 4\pi \int_{\tau_{k+1}}^{\tau_k} J_\theta(\tau_k - u)\langle \varphi, P_{2u}\varphi\rangle_{L^2}\,\mathrm{d}u.\end{aligned}$$

Since the heat semigroup is an L^2 contraction, $|\langle \varphi, P_{2u}\varphi\rangle| \le \|\varphi\|_2^2$; the asymptotics of J_θ on this one slab then give

$$\mathbb{E}|\langle G_k - 1, \varphi\rangle|^2 \le C\frac{\tau_k}{k}\|\varphi\|_2^2.$$

For the nonlinear remainder, the previously proved pointwise bound $\|T_k(y)\|_2 \le C/k$ is enough. Minkowski's integral inequality gives

$$\begin{aligned}\|\langle T_k - \mathbb{E}T_k, \varphi\rangle\|_2 &\le \int |\varphi(y)|\;\|T_k(y) - \mathbb{E}T_k(y)\|_2\;\mathrm{d}y \\ &\le \frac{C}{k}\|\varphi\|_1,\end{aligned}$$

so

$$\mathbb{E}|\langle T_k - \mathbb{E}T_k, \varphi\rangle|^2 \le Ck^{-2}\|\varphi\|_1^2.$$

This is the precise role of Minkowski here: it converts a uniform pointwise $L^2(\Omega)$ estimate into an $L^2(\Omega)$ estimate after spatial testing without requiring any spatial covariance information for T_k. Different k use independent slabs. Summing the variances of the centered logarithmic pairings gives

$$\mathbb{E}\left|\sum_{k=m}^{n-1}\langle \log G_k - \mathbb{E}\log G_k, \varphi\rangle\right|^2 \le \frac{C}{m}\big(\|\varphi\|_1^2 + \tau_m\|\varphi\|_2^2\big).$$

The deterministic mean is handled by (85); the actual composition error is handled by (86). These prove (87). □

5.2 From moving microscopic times to an original fixed-time field

The aligned multiscale construction controls a window whose endpoints move toward time zero. The original regularized field, however, is defined on a fixed macroscopic time interval. This section separates these two issues: first compare the moving fine window with a field carrying an additional fixed coarse interval, and only afterward fill the gaps between the geometric Gaussian scales.

Fix $T > 0$ and write

$$F_a^T(y) = p_a(\cdot - y) \triangleleft Z_{0,T}^\theta \triangleright 1, \qquad a > 0.$$

Choose a fixed sufficiently large M such that $\tau_M < T$, and let $n > M$. On the same SHF define

$$Y_{M,n}(y) = p_{\tau_n}(\cdot - y) \triangleleft Z_{\tau_n,\tau_M}^\theta \triangleright 1, \qquad B_n(y) = p_{\tau_n}(\cdot - y) \triangleleft Z_{\tau_n,T+\tau_n}^\theta \triangleright 1.$$

Time stationarity implies that B_n and $F_{\tau_n}^T$ have the same law as spatial fields. This equality is used only for marginal bounds; it is not a common-noise coupling across n.

The following comparison removes the moving-time restriction of the aligned construction. It shows that attaching the fixed coarse time interval changes the logarithm by only a uniformly L^1-bounded amount and identifies the fixed-time mean to the required accuracy.

Lemma 5.2 (A compactness transfer through the coarse time interval)**.** *With $D_n = \log(B_n/A_n)$,*

$$\sup_{n>M} \sup_y \mathbb{E}|D_n(y)| \le C_{M,T,\theta,b}. \tag{94}$$

Moreover

$$\mathbb{E} \log F_{\tau_n}^T(y) = -\frac{1}{2} \log n + O_{M,T,\theta,b}(1). \tag{95}$$

Proof. Let $\mathcal{A}_n$ be the sigma-field of the SHF between τ_n and τ_M. The mean heat kernel and terminal test 1 give

$$\mathbb{E}[B_n(y) \mid \mathcal{A}_n] = A_n(y), \qquad \mathbb{E}[B_n(y)/A_n(y) \mid \mathcal{A}_n] = 1.$$

Then Jensen gives $\mathbb{E} \log B_n \le \mathbb{E} \log A_n$, and

$$\mathbb{E}[(D_n)_+ \mid \mathcal{A}_n] = \mathbb{E}[(\log(B_n/A_n))_+ \mid \mathcal{A}_n] \le \mathbb{E}[B_n/A_n \mid \mathcal{A}_n] = 1,$$

which implies $\mathbb{E}(D_n)_+ \le 1$.

By (75),

$$\mathbb{E}B_n^2 \le 1 + J_\theta(T) \log(1 + T/\tau_n).$$

Since $\tau_n = b^{2n}$, we have

$$\log(1 + T/\tau_n) = \log(1 + Tb^{-2n}) \le C_{T,b} + 2n \log b^{-1} \le C_{T,b} n,$$

for $n > M$. Hence

$$\mathbb{E}B_n^2 \le C_{T,\theta,b} n. \tag{96}$$

Moreover, the mean heat kernel and the terminal test 1 give $\mathbb{E}B_n = 1$. Therefore Proposition 4.2, applied to B_n, yields

$$\mathbb{E} \log B_n = \mathbb{E} \log(B_n/\mathbb{E}B_n) \ge -\frac{1}{2} \log \frac{\mathbb{E}B_n^2}{(\mathbb{E}B_n)^2} = -\frac{1}{2} \log \mathbb{E}B_n^2.$$

Using (96),

$$\mathbb{E} \log B_n \ge -\frac{1}{2} \log(C_{T,\theta,b} n) \ge -\frac{1}{2} \log n - C_{T,\theta,b}. \tag{97}$$

We next obtain the corresponding upper bound for $\mathbb{E} \log A_n$. Since $A_n = Y_{M,n}$, the definition (5) gives

$$\log A_n = \sum_{k=M}^{n-1} \log G_k + \Lambda_{M,n}.$$

Taking expectations,

$$\mathbb{E}\log A_n = \sum_{k=M}^{n-1} \mathbb{E}\log G_k + \mathbb{E}\Lambda_{M,n}.$$

For every real number x one has $x \le x_+$, so

$$\mathbb{E}\Lambda_{M,n} \le \mathbb{E}(\Lambda_{M,n})_+.$$

The one-sided estimate (61), with $m = M$, therefore implies

$$\mathbb{E}\Lambda_{M,n} \le CM^{-1/2}.$$

On the other hand, (85) gives, uniformly over $n > M$,

$$\sum_{k=M}^{n-1} \mathbb{E}\log G_k = -\frac{1}{2}\log(n/M) + O(M^{-1/2}).$$

Combining the preceding two estimates, we obtain

$$\mathbb{E}\log A_n \le -\frac{1}{2}\log(n/M) + \frac{C}{\sqrt{M}}. \tag{98}$$

Recall that

$$D_n = \log(B_n/A_n) = \log B_n - \log A_n.$$

Hence (97) and (98) imply

$$\begin{aligned}\mathbb{E}D_n &= \mathbb{E}\log B_n - \mathbb{E}\log A_n \\ &\ge -\frac{1}{2}\log n - C + \frac{1}{2}\log(n/M) - \frac{C}{\sqrt{M}} \\ &= -\frac{1}{2}\log M - C - \frac{C}{\sqrt{M}}.\end{aligned}$$

Since M is fixed and sufficiently large, the last term can be absorbed into the constant, and therefore

$$\mathbb{E}D_n \ge -C - \frac{1}{2}\log M. \tag{99}$$

From the conditional-mean identity established above we already know that $\mathbb{E}(D_n)_+ \le 1$. For every real number x,

$$|x| = 2x_+ - x,$$

and consequently

$$\mathbb{E}|D_n| = 2\mathbb{E}(D_n)_+ - \mathbb{E}D_n.$$

Using $\mathbb{E}(D_n)_+ \le 1$ together with (99), we find

$$\mathbb{E}|D_n| \le 2 + C + \frac{1}{2}\log M.$$

The bound is uniform in $n > M$ and in y, so this proves (94).

It remains to record the mean asymptotic (95). The lower bound is already contained in (97):

$$\mathbb{E}\log B_n \ge -\frac{1}{2}\log n - C.$$

For the upper bound, the conditional Jensen inequality obtained above gives

$$\mathbb{E}\log B_n \le \mathbb{E}\log A_n.$$

Applying (98),

$$\mathbb{E}\log B_n \le -\frac{1}{2}\log(n/M) + \frac{C}{\sqrt{M}} = -\frac{1}{2}\log n + \frac{1}{2}\log M + \frac{C}{\sqrt{M}}.$$

Thus, for fixed M,

$$\mathbb{E}\log B_n = -\frac{1}{2}\log n + O_{M,T,\theta,b}(1).$$

Finally, time stationarity implies that B_n and $F^T_{\tau_n}$ have the same law, and hence

$$\mathbb{E}\log F^T_{\tau_n}(y) = -\frac{1}{2}\log n + O_{M,T,\theta,b}(1),$$

which is precisely (95). □

Combining (87) with (94), we now pass from the fine-scale field A_n to the original fixed-time field. Since

$$D_n = \log B_n - \log A_n,$$

we may write

$$\log B_n + \frac{1}{2}\log n = \left(\log A_n + \frac{1}{2}\log(n/M)\right) + D_n + \frac{1}{2}\log M.$$

Pairing this identity against a test function φ gives

$$\left\langle \log B_n + \frac{1}{2}\log n, \varphi \right\rangle = \left\langle \log A_n + \frac{1}{2}\log(n/M), \varphi \right\rangle + \langle D_n, \varphi\rangle + \frac{1}{2}\log M \int_{\mathbb{R}^2} \varphi(y)\,dy.$$

Hence, by the triangle inequality,

$$\begin{aligned}\mathbb{E}\left|\left\langle \log B_n + \frac{1}{2}\log n, \varphi \right\rangle\right| \le{}& \mathbb{E}\left|\left\langle \log A_n + \frac{1}{2}\log(n/M), \varphi \right\rangle\right| \\ &+ \mathbb{E}|\langle D_n, \varphi\rangle| + \frac{1}{2}\log M \left|\int_{\mathbb{R}^2} \varphi(y)\,dy\right|.\end{aligned}$$

We estimate the three terms on the right-hand side separately. First, since $A_n = Y_{M,n}$, (87) with $m = M$ yields

$$\mathbb{E}\left|\left\langle \log A_n + \frac{1}{2}\log(n/M), \varphi \right\rangle\right| \le \frac{C}{\sqrt{M}}\big(\|\varphi\|_1 + \|\varphi\|_2\big).$$

Since M is fixed, the factor $M^{-1/2}$ may be absorbed into the constant.

For the second term,

$$|\langle D_n, \varphi\rangle| = \left|\int_{\mathbb{R}^2} D_n(y)\varphi(y)\,dy\right| \le \int_{\mathbb{R}^2} |D_n(y)|\,|\varphi(y)|\,dy.$$

Therefore Tonelli's theorem and (94) imply

$$\begin{aligned}\mathbb{E}|\langle D_n, \varphi\rangle| &\le \int_{\mathbb{R}^2} |\varphi(y)|\, \mathbb{E}|D_n(y)|\, dy \\ &\le \left(\sup_{n>M}\sup_{y\in\mathbb{R}^2} \mathbb{E}|D_n(y)|\right) \|\varphi\|_1 \\ &\le C_{M,T,\theta,b}\, \|\varphi\|_1 .\end{aligned}$$

Finally,

$$\frac{1}{2}\log M \left|\int_{\mathbb{R}^2} \varphi(y)\, dy\right| \le \frac{1}{2}\log M\ \|\varphi\|_1 .$$

Combining these estimates, and enlarging the constant if necessary, we obtain

$$\sup_{n>M} \mathbb{E}\left|\left\langle \log B_n + \frac{1}{2}\log n, \varphi\right\rangle\right| \le C\big(\|\varphi\|_1 + \|\varphi\|_2\big),$$

where C depends on M, T, θ, b but not on n.

By time stationarity, B_n and $F^T_{\tau_n}$ have the same law as spatial fields. Hence

$$\left\langle \log B_n + \frac{1}{2}\log n, \varphi\right\rangle \overset{d}{=} \left\langle \log F^T_{\tau_n} + \frac{1}{2}\log n, \varphi\right\rangle,$$

and therefore

$$\sup_{n>M} \mathbb{E}\left|\left\langle \log F^T_{\tau_n} + \frac{1}{2}\log n, \varphi\right\rangle\right| \le C\big(\|\varphi\|_1 + \|\varphi\|_2\big). \tag{100}$$

All Gaussian scales, not just a geometric subsequence. This step is essential because the target family is indexed by every $\varepsilon \downarrow 0$, not only by the geometric sequence $\varepsilon = b^n$. Tightness on a subsequence of scales would not imply tightness of the full family.

Given $a \in [\tau_{n+1}, \tau_n]$, we have $a \le \tau_n$, and hence

$$\frac{p_a(x)}{p_{\tau_n}(x)} = \frac{\tau_n}{a}\exp\left\{-\frac{|x|^2}{2}\left(\frac{1}{a} - \frac{1}{\tau_n}\right)\right\} \le \frac{\tau_n}{a} \le b^{-2}.$$

Thus

$$p_a(x) \le \frac{\tau_n}{a} p_{\tau_n}(x) \le b^{-2} p_{\tau_n}(x).$$

Because the SHF kernel is a positive measure, integrating this deterministic pointwise inequality against $Z^\theta_{0,T}$ preserves the order:

$$F^T_a(y) \le b^{-2} F^T_{\tau_n}(y).$$

Taking logarithms gives $\log F^T_a(y) - \log F^T_{\tau_n}(y) \le 2L$. The second-moment formula and Gaussian convexity give

$$\mathbb{E}\log F^T_a \ge -\tfrac{1}{2}\log n - C,$$

whereas (95) gives the matching upper bound for $\mathbb{E}\log F^T_{\tau_n}$. Hence $\mathbb{E}\log F^T_a(y) - \log F^T_{\tau_n}(y) \ge -C$ and $\mathbb{E}|\log F^T_a(y) - \log F^T_{\tau_n}(y)| \le 4L + C$, uniformly in a, n, y. Since $\log(1 + \log a^{-1}) - \log n$ is bounded on these intervals, (100) extends to

$$\sup_{0<\varepsilon<\varepsilon_0} \mathbb{E}\left|\big\langle \log F^T_{\varepsilon^2} + c_\varepsilon, \varphi\big\rangle\right| \le C\big(\,\|\varphi\|_1 + \|\varphi\|_2\,\big). \tag{101}$$

We also have the uniform pointwise mean asymptotic

$$\mathbb{E}\log F^T_{\varepsilon^2}(y) = -c_\varepsilon + O(1). \tag{102}$$

5.3 General initial data and each fixed mollifier

We use the time-reversal/transposition invariance of the SHF on a fixed time interval; Gu–Tsai explicitly invoke this invariance in Section 1.5 of [6] to move a terminal heat-kernel test to the starting time. In measure notation, together with time stationarity, it gives

$$Z^\theta_{s,t}(\,\mathrm{d}x,\,\mathrm{d}z) \overset{\text{law}}{=} Z^\theta_{0,T}(\,\mathrm{d}z,\,\mathrm{d}x), \qquad T = t - s,$$

for the tested finite-dimensional distributions used below. Therefore, with $\check{\rho}(x) = \rho(-x)$,

$$\begin{aligned} F^\rho_\varepsilon(y) &= \iint g(x)\rho_\varepsilon(y-z)Z^\theta_{s,t}(\,\mathrm{d}x,\,\mathrm{d}z) \\ &\overset{\text{law}}{=} \iint \check{\rho}_\varepsilon(x-y)g(z)Z^\theta_{0,T}(\,\mathrm{d}x,\,\mathrm{d}z) =: \widehat{F}_\varepsilon(y). \end{aligned}$$

Since $\check{\rho}$ is again admissible, we relabel $\check{\rho}$ as ρ for the rest of this section. Thus no symmetry assumption on the original mollifier is needed.

Because the resulting fixed profile ρ is compactly supported, some $C_\rho < \infty$ satisfies $\rho(x) \le C_\rho p_1(x)$ for all x: on the compact support of ρ, the strictly positive function p_1 has a positive minimum, while outside the support the inequality is trivial. Scaling gives

$$\rho_\varepsilon(x-y) \le C_\rho p_{\varepsilon^2}(x-y).$$

Since $0 \le g \le \|g\|_\infty$ and the SHF is a positive measure, this pointwise kernel comparison passes directly through the double integral. On the same SHF,

$$0 < \widehat{F}_\varepsilon(y) \le C_\rho \,\|g\|_\infty F^T_{\varepsilon^2}(y). \tag{103}$$

Note that

$$\mathbb{E}\widehat{F}_\varepsilon(y) = \int_{\mathbb{R}^2} \rho_\varepsilon(x-y)(P_T g)(x)\,\mathrm{d}x.$$

Since g is positive on some ball, there exist a point $z_0 \in \mathbb{R}^2$, a radius $r > 0$, and a constant $a > 0$ such that

$$g(z) \ge a \qquad \text{for } z \in B(z_0, r).$$

Therefore

$$(P_T g)(x) = \int_{\mathbb{R}^2} p_T(x-z)g(z)\,\mathrm{d}z \ge a \int_{B(z_0,r)} p_T(x-z)\,\mathrm{d}z.$$

Using the Gaussian heat kernel bound

$$p_T(x-z) = \frac{1}{2\pi T}\exp\!\left(-\frac{|x-z|^2}{2T}\right),$$

and the fact that z ranges over a fixed bounded set, we obtain

$$(P_T g)(x) \ge c\exp\{-C(1+|x|^2)\}.$$

Now suppose that $\rho_\varepsilon(x-y) \ne 0$. Since ρ is compactly supported, there is a fixed $R < \infty$ such that, for all sufficiently small ε,

$$|x-y| \le R.$$

Hence

$$1 + |x|^2 \le C(1+|y|^2)$$

uniformly on the support of $\rho_\varepsilon(x-y)$. Consequently

$$(P_T g)(x) \geq c \exp\{-C(1+|y|^2)\}$$

whenever $\rho_\varepsilon(x-y) \neq 0$. Since ρ_ε has total mass one, it follows that, uniformly for all sufficiently small ε,

$$\mathbb{E}\widehat{F}_\varepsilon(y) = \int \rho_\varepsilon(x-y)(P_T g)(x)\, \mathrm{d}x \geq c \exp\{-C(1+|y|^2)\}. \tag{104}$$

We next control the second moment. By (103),

$$0 < \widehat{F}_\varepsilon(y) \leq C_\rho \, \|g\|_\infty \, F^T_{\varepsilon^2}(y), \tag{105}$$

and therefore

$$\mathbb{E}\widehat{F}_\varepsilon(y)^2 \leq C\, \mathbb{E}\big(F^T_{\varepsilon^2}(y)\big)^2.$$

Applying (75) to the Gaussian-start field yields

$$\mathbb{E}\big(F^T_{\varepsilon^2}(y)\big)^2 \leq C\big(1+\log \varepsilon^{-2}\big),$$

and hence

$$\mathbb{E}\widehat{F}_\varepsilon(y)^2 \leq C\big(1+\log \varepsilon^{-2}\big). \tag{106}$$

Proposition 4.2 gives

$$\begin{aligned}\mathbb{E}\log \widehat{F}_\varepsilon(y) &\geq \log \mathbb{E}\widehat{F}_\varepsilon(y) - \frac{1}{2}\log \frac{\mathbb{E}\widehat{F}_\varepsilon(y)^2}{|\mathbb{E}\widehat{F}_\varepsilon(y)|^2} \\ &= 2\log \mathbb{E}\widehat{F}_\varepsilon(y) - \frac{1}{2}\log \mathbb{E}\widehat{F}_\varepsilon(y)^2.\end{aligned}$$

By (104),

$$\log \mathbb{E}\widehat{F}_\varepsilon(y) \geq -C(1+|y|^2),$$

while (106) implies

$$-\frac{1}{2}\log \mathbb{E}\widehat{F}_\varepsilon(y)^2 \geq -\frac{1}{2}\log\big(C(1+\log\varepsilon^{-2})\big).$$

Absorbing the constant factor into the additive constant and using the definition of the centering c_ε, we conclude that

$$\mathbb{E}\log \widehat{F}_\varepsilon(y) \geq -c_\varepsilon - C(1+|y|^2). \tag{107}$$

Now we estimate $\log \widehat{F}_\varepsilon(y)$ by

$$\log F^T_{\varepsilon^2}(y) = p_{\varepsilon^2}(\cdot - y) \triangleleft Z^\theta_{0,T} \triangleright 1$$

The domination estimate (105) gives the pointwise bound for the error

$$\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y) \leq \log\big(C_\rho \, \|g\|_\infty\big),$$

and therefore

$$(\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y))_+ \leq \max\{0, \log(C_\rho \, \|g\|_\infty)\}.$$

Thus the positive part of $V_\varepsilon(y)$ is bounded deterministically, uniformly in ε and y.

On the other hand, by (107) and the Gaussian mean estimate (102),

$$\mathbb{E}\log \widehat{F}_\varepsilon(y) \geq -c_\varepsilon - C(1+|y|^2),$$

whereas

$$\mathbb{E}\log F^T_{\varepsilon^2}(y) = -c_\varepsilon + O(1).$$

The singular centering c_ε therefore cancels, and we obtain

$$\mathbb{E}[\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y)] \geq -C(1+|y|^2).$$

For every real number x one has

$$|x| = 2x_+ - x.$$

Hence

$$\mathbb{E}|\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y)| = 2\mathbb{E}(\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y))_+ - \mathbb{E}[\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y)].$$

Using the deterministic bound on $(V_\varepsilon(y))_+$ and the preceding lower bound on $\mathbb{E}V_\varepsilon(y)$, we arrive at

$$\boxed{\mathbb{E}|\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y)| \leq C(1+|y|^2).} \tag{108}$$

Therefore, for every test function φ,

$$\mathbb{E}\left|\left\langle \log \widehat{F}_\varepsilon + c_\varepsilon, \varphi \right\rangle\right| \leq \mathbb{E}\left|\left\langle \log F^T_{\varepsilon^2} + c_\varepsilon, \varphi\right\rangle\right| + \mathbb{E}|\langle \log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y), \varphi\rangle|.$$

The first term is controlled uniformly in ε by (101). For the second term, Tonelli's theorem and (108) give

$$\begin{aligned}\mathbb{E}|\langle \log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y), \varphi\rangle| &\leq \int_{\mathbb{R}^2} |\varphi(y)|\, \mathbb{E}|\log \widehat{F}_\varepsilon(y) - \log F^T_{\varepsilon^2}(y)|\, \mathrm{d}y \\ &\leq C\int_{\mathbb{R}^2} (1+|y|^2)|\varphi(y)|\, \mathrm{d}y.\end{aligned}$$

Since φ is compactly supported, the right-hand side is controlled by the test-function seminorm appearing in (2). Combining this with (101) proves (2).

6 Tightness of regularized logarithms

We now upgrade the scalar estimate (2) directly to tightness of the full random distributions. We use the static nuclear-space criterion for characteristic functionals; see [1, Corollary 4.6.8]. We also use the fact that $\mathcal{D}'_\beta(\mathbb{R}^2)$ is a Prohorov space.

Set

$$\mathcal{H}_\varepsilon := H^{g,\rho}_\varepsilon + c_\varepsilon, \qquad \mu_\varepsilon := \mathrm{Law}(\mathcal{H}_\varepsilon).$$

As verified above, $\mathcal{H}_\varepsilon$ is a $\mathcal{D}'_\beta(\mathbb{R}^2)$-valued random variable. Its law is Radon: indeed, $\mathcal{H}_\varepsilon$ is obtained from a $C^\infty_{\mathrm{loc}}(\mathbb{R}^2)$-valued random variable through the continuous canonical embedding

$$C^\infty_{\mathrm{loc}}(\mathbb{R}^2) \hookrightarrow \mathcal{D}'_\beta(\mathbb{R}^2),$$

and $C^\infty_{\mathrm{loc}}(\mathbb{R}^2)$ is a Polish space.

For $\varphi \in \mathcal{D}(\mathbb{R}^2)$, let

$$\widehat{\mu}_\varepsilon(\varphi) := \int_{\mathcal{D}'_\beta} e^{i\langle u,\varphi\rangle}\, \mu_\varepsilon(\mathrm{d}u) = \mathbb{E}e^{i\langle \mathcal{H}_\varepsilon,\varphi\rangle}$$

be the characteristic functional.

Recall from Theorem 1.1 that

$$\sup_{0<\varepsilon<\varepsilon_0} \mathbb{E}|\langle \mathcal{H}_\varepsilon, \varphi\rangle| \le Cq(\varphi), \tag{109}$$

where

$$q(\varphi) = \|\varphi\|_{L^2} + \int_{\mathbb{R}^2} (1+|y|^2)|\varphi(y)|\,\mathrm{d}y.$$

The seminorm q is continuous on $\mathcal{D}(\mathbb{R}^2)$ with its usual strict LF topology. Indeed, if $\operatorname{supp}\varphi \subset K$, then

$$q(\varphi) \le \left(|K|^{1/2} + \int_K (1+|y|^2)\,\mathrm{d}y \right) \|\varphi\|_\infty,$$

so the restriction of q to every $\mathcal{D}_K$ is continuous.

Using $|e^{ix}-1| \le |x|$ and (109), we obtain

$$\begin{aligned} |\widehat{\mu}_\varepsilon(\varphi) - 1| &\le \mathbb{E}\left| e^{i\langle \mathcal{H}_\varepsilon, \varphi\rangle} - 1 \right| \\ &\le \mathbb{E}|\langle \mathcal{H}_\varepsilon, \varphi\rangle| \\ &\le Cq(\varphi). \end{aligned}$$

Consequently,

$$\sup_{0<\varepsilon<\varepsilon_0} |\widehat{\mu}_\varepsilon(\varphi) - 1| \longrightarrow 0 \qquad \text{as } \varphi \longrightarrow 0 \text{ in } \mathcal{D}(\mathbb{R}^2).$$

Thus the family of characteristic functionals $\{\widehat{\mu}_\varepsilon : 0<\varepsilon<\varepsilon_0\}$ is equicontinuous at the origin.

Bogachev's collorary [1, Corollary 4.6.8] applies to $D'_\beta(\mathbb{R}^2)$ as stated in their book, so $\{\mu_\varepsilon : 0 < \varepsilon < \varepsilon_0\}$ has compact closure in the weak topology of probability measures on $\mathcal{D}'_\beta(\mathbb{R}^2)$. Moreover, $\mathcal{D}'_\beta(\mathbb{R}^2)$ is a Prohorov space. Hence every weakly compact family of Radon probability measures on $\mathcal{D}'_\beta(\mathbb{R}^2)$ is uniformly tight. We conclude that

$$\boxed{\{\mathcal{H}_\varepsilon : 0<\varepsilon<\varepsilon_0\} \text{ is uniformly tight in } \mathcal{D}'_\beta(\mathbb{R}^2).} \tag{110}$$

Finally, let

$$Q : \mathcal{D}'_\beta \longrightarrow \mathcal{D}'_\beta / \operatorname{span}\{1\} \simeq \mathcal{D}'_{0,\beta}$$

be the continuous quotient map. Since c_ε is spatially constant,

$$Q(\mathcal{H}_\varepsilon) = [H_\varepsilon^{g,\rho}].$$

Continuous images of uniformly tight families are uniformly tight. Therefore

$$\boxed{\{[H_\varepsilon^{g,\rho}] : 0<\varepsilon<\varepsilon_0\} \text{ is uniformly tight in } \mathcal{D}'_{0,\beta}.}$$

This proves Corollary 1.2.

A External results used in the proof

This appendix records the non-elementary external inputs in the forms used in the body. The statements are adapted to the present notation; the cited sources contain the precise original hypotheses and normalizations. Standard inequalities and convergence theorems are not repeated.

A.1 Structural and approximation results for the SHF

The first input fixes the probabilistic structure that makes the slab factorizations and conditional expectations in the proof meaningful.

Proposition A.1 (SHF structure; Caravenna–Sun–Zygouras and Tsai)**.** *For each $\theta \in \mathbb{R}$, the critical two-dimensional SHF is a family of positive locally finite random measure kernels $Z^{\theta}_{s,t}(\,\mathrm{d}x,\,\mathrm{d}z)$, $s < t$, with the mean heat kernel, spatial and time stationarity, independence on disjoint time slabs, and the stochastic-flow composition property. We also use the time-reversal/transposition invariance in the form invoked in Gu–Tsai [6, Section 1.5]. These properties are part of the SHF construction and characterization in [2, 5].*

The next approximation theorem is used only to transfer a dimension-free Gaussian inequality to a limiting SHF mass and to justify fixed-time moment bounds.

Proposition A.2 (Gaussian approximation and moment convergence; Tsai)**.** *Let $F = f \triangleleft Z^{\theta}_{u,v} \triangleright h$ with nonnegative compact smooth deterministic profiles. Tsai's noise-mollified SHE approximation produces positive Gaussian-functionals F_η converging in law to F, with convergence of the first and second moments. The moment-measure formulas also give the fixed-time higher moments used in this paper. We use [5, Proposition 1.5, Proposition 1.7, Corollary 1.8]. Heat profiles and the terminal test 1 are obtained by monotone approximation.*

A.2 Local multiscale estimates of Gu–Tsai

The next input is the only microscopic signed-chain estimate that is not proved from the two-replica formula in this paper. Its role is to control one fixed error interval before the all-length summation.

Proposition A.3 (Single-chain and one-block bounds; Gu–Tsai)**.** *After the geometric-index reversal described in Lemma 2.3, Gu–Tsai's Corollary 3.5 yields, for $J = [j, j + h - 1]$ with $h \geq 2$,*

$$\|A_J(y)\|_2 \leq \frac{C\sqrt{L}}{j} \left(\frac{C_0}{L}\right)^{h/2},$$

uniformly in the spatial center. Their Corollary B.3 gives the one-block high-moment estimate

$$\mathbb{E}|G_k(y) - 1|^r \leq C_r k^{-r/2}$$

in the range supplied by that corollary; the finitely many remaining indices are covered here using Tsai's fixed-time moment formulas. See [6, Corollary 3.5, Corollary B.3].

The two-replica input is used to compute exact second moments and spatial covariances of the one-block fields.

Proposition A.4 (Two-replica moment kernel; Tsai and Gu–Tsai)**.** *Let $Z^{\theta}_{0,t}$ be the stochastic heat flow with parameter θ. For nonnegative deterministic profiles f, h, write*

$$f \triangleleft Z^{\theta}_{0,t} \triangleright h := \iint_{\mathbb{R}^2 \times \mathbb{R}^2} f(x) h(z)\, Z^{\theta}_{0,t}(\,\mathrm{d}x,\,\mathrm{d}z).$$

Let

$$p_t(x) = \frac{1}{2\pi t} \exp\left(-\frac{|x|^2}{2t}\right), \qquad t > 0,\ x \in \mathbb{R}^2,$$

and define

$$\mathsf{j}^{\theta}(r)=\int_0^{\infty}\frac{r^{\alpha-1}e^{\theta\alpha}}{\Gamma(\alpha)}\,\mathrm{d}\alpha,\qquad J_{\theta}(s)=\int_0^{s}\mathsf{j}^{\theta}(r)\,\mathrm{d}r.$$

Then, for every $a,t>0$ and $y,z\in\mathbb{R}^2$,

$$\begin{aligned}\mathbb{E}\Big[p_a(\cdot-y)\triangleleft Z^{\theta}_{0,t}\triangleright 1\cdot p_a(\cdot-z)\triangleleft Z^{\theta}_{0,t}\triangleright 1\Big]&\\ =1+4\pi\int_{0<u<v<t}\mathsf{j}^{\theta}(v-u)\,p_{2(a+u)}(y-z)\,\mathrm{d}u\,\mathrm{d}v&\\ =1+4\pi\int_0^t p_{2(a+u)}(y-z)J_{\theta}(t-u)\,\mathrm{d}u.&\end{aligned}\tag{111}$$

In particular, setting $z=y$,

$$\mathbb{E}\left|p_a(\cdot-y)\triangleleft Z^{\theta}_{0,t}\triangleright 1\right|^2=1+\int_0^t\frac{J_{\theta}(t-u)}{a+u}\,\mathrm{d}u.\tag{112}$$

Proof. Tsai's moment identity, specialized to $n=2$, identifies the second moment of tested SHF masses with the two-particle delta-Bose semigroup:

$$\mathbb{E}\Big[Z^{\theta}_{0,t}(g_1\otimes h_1)\,Z^{\theta}_{0,t}(g_2\otimes h_2)\Big]=\Big\langle g_1\otimes g_2,\,\mathcal{Q}^{[[2]],\theta}(t)(h_1\otimes h_2)\Big\rangle.$$

Since the terminal profile 1 is not in $L^2(\mathbb{R}^2)$, we first choose nonnegative compactly supported cutoffs $\chi_R\uparrow 1$, apply the identity with

$$g_1=p_a(\cdot-y),\qquad g_2=p_a(\cdot-z),\qquad h_1=h_2=\chi_R,$$

and then let $R\to\infty$ by positivity and monotone convergence.

The explicit two-replica kernel decomposes $\mathcal{Q}^{[[2]],\theta}(t)$ into the free two-particle heat semigroup and an interaction term. After integration against the terminal profile 1, each free replica has total mass one, so the free contribution is 1.

For the interaction term, the explicit kernel contributes

$$4\pi\,\mathsf{j}^{\theta}(v-u),\qquad 0<u<v<t.$$

The two terminal heat kernels disappear after integration against the terminal profile, since

$$\int_{\mathbb{R}^2}p_{t-v}(w-x)\,\mathrm{d}w=1.$$

On the starting side, the first replica contributes

$$\int_{\mathbb{R}^2}p_a(w-y)p_u(x-w)\,\mathrm{d}w=(p_a*p_u)(x-y)=p_{a+u}(x-y),$$

while the second contributes $p_{a+u}(x-z)$. Hence the two starting legs give

$$\int_{\mathbb{R}^2}p_{a+u}(x-y)p_{a+u}(x-z)\,\mathrm{d}x.$$

Using symmetry of the heat kernel and the semigroup property,

$$\int_{\mathbb{R}^2}p_{a+u}(x-y)p_{a+u}(x-z)\,\mathrm{d}x=(p_{a+u}*p_{a+u})(y-z)=p_{2(a+u)}(y-z).$$

Therefore

$$\mathbb{E}\Big[p_a(\cdot - y) \triangleleft Z^\theta_{0,t} \triangleright 1 p_a(\cdot - z) \triangleleft Z^\theta_{0,t} \triangleright 1\Big] = 1 + 4\pi \int_{0<u<v<t} \mathsf{j}^\theta(v-u) p_{2(a+u)}(y-z)\, \mathrm{d}u\, \mathrm{d}v.$$

For fixed u,

$$\int_u^t \mathsf{j}^\theta(v-u)\, \mathrm{d}v = \int_0^{t-u} \mathsf{j}^\theta(r)\, \mathrm{d}r = J_\theta(t-u),$$

which proves (111).

The diagonal formula is now immediate. Setting $z = y$ gives

$$p_{2(a+u)}(0) = \frac{1}{4\pi(a+u)},$$

and hence

$$\mathbb{E}\left|p_a(\cdot - y) \triangleleft Z^\theta_{0,t} \triangleright 1\right|^2 = 1 + \int_0^t \frac{J_\theta(t-u)}{a+u}\, \mathrm{d}u,$$

which is (112).

□

A.3 Gaussian convexity

This theorem is the source of the inverse-moment and free-energy lower bounds used to control negative logarithmic spikes.

Theorem A.5 (Chen's Gaussian convexity theorem)**.** *Let g be a finite-dimensional standard Gaussian vector and let $f(g)$ be a convex function such that $\mathbb{E}e^{uf(g)} < \infty$ for every $u > 0$. Then*

$$\Phi(u) = \frac{1}{u} \log \mathbb{E}e^{uf(g)}, \qquad \Phi(0) = \mathbb{E}f(g),$$

is convex as a function of $u \in \mathbb{R}$. This is [4, Theorem 1].

A.4 Tightness in the strong dual of a nuclear LF space

We record the static compactness criterion used in Section 6. The formulation below is adapted from Bogachev [1, Corollary 4.6.8].

Theorem A.6 (Bogachev's nuclear-space compactness criterion)**.** *Let X be a reflexive nuclear locally convex space, and let*

$$X^*$$

denote its continuous dual. Equip X^* with the strong dual topology

$$\beta(X^*, X),$$

that is, the topology of uniform convergence on bounded subsets of X. Let $\mathcal{M}$ be a family of Radon probability measures on

$$\big(X^*, \beta(X^*, X)\big).$$

For $\mu \in \mathcal{M}$, define its characteristic functional by

$$\widehat{\mu}(x) := \int_{X^*} e^{\,i\langle u, x\rangle}\, \mu(\mathrm{d}u), \qquad x \in X,$$

where $\langle u, x\rangle$ denotes the canonical dual pairing.

Assume that the family of characteristic functionals

$$\{\widehat{\mu} : \mu \in \mathcal{M}\}$$

is equicontinuous at the origin of X, i.e. for every $\delta > 0$ there exists a neighborhood U of 0 in X such that

$$\sup_{\mu\in\mathcal{M}} |\widehat{\mu}(x) - 1| < \delta \qquad \text{for every } x \in U.$$

Then $\mathcal{M}$ has compact closure in the weak topology of probability measures on $\big(X^*, \beta(X^*, X)\big)$.

Bogachev explicitly notes that this corollary applies, in particular, to

$$X^* = \mathcal{S}'(\mathbb{R}^d) \qquad \text{and} \qquad X^* = \mathcal{D}'(\mathbb{R}^d).$$

Remark A.7 (Dictionary with the notation of the present paper). In our application,

$$X = \mathcal{D}(\mathbb{R}^2) = C_c^\infty(\mathbb{R}^2),$$

equipped with its usual strict LF topology. Its continuous dual is

$$X^* = \mathcal{D}'(\mathbb{R}^2),$$

and the strong dual topology used by Bogachev is exactly the topology denoted throughout this paper by

$$\mathcal{D}'_\beta(\mathbb{R}^2) = \big(\mathcal{D}'(\mathbb{R}^2), \beta(\mathcal{D}'(\mathbb{R}^2), \mathcal{D}(\mathbb{R}^2))\big).$$

Thus Bogachev's pair

$$\big(X^*, \beta(X^*, X)\big)$$

is precisely our strong distribution space

$$\mathcal{D}'_\beta(\mathbb{R}^2).$$

For each $\varepsilon > 0$ we set, as in Section 6,

$$\mathcal{H}_\varepsilon := H_\varepsilon^{g,\rho} + c_\varepsilon,$$

and denote its law on $\mathcal{D}'_\beta(\mathbb{R}^2)$ by

$$\mu_\varepsilon := \mathrm{Law}(\mathcal{H}_\varepsilon).$$

Accordingly, Bogachev's family $\mathcal{M}$ is taken to be

$$\mathcal{M} = \{\mu_\varepsilon : 0 < \varepsilon < \varepsilon_0\}.$$

For $\varphi \in \mathcal{D}(\mathbb{R}^2)$, the characteristic functional appearing in Bogachev's theorem becomes

$$\widehat{\mu}_\varepsilon(\varphi) = \int_{\mathcal{D}'_\beta} e^{\,i\langle u,\varphi\rangle}\, \mu_\varepsilon(\mathrm{d}u) = \mathbb{E} e^{\,i\langle \mathcal{H}_\varepsilon,\varphi\rangle}.$$

Hence the abstract dual pairing

$$\langle u, x\rangle, \qquad u \in X^*,\ x \in X,$$

is exactly the distribution–test-function pairing

$$\langle u, \varphi\rangle, \qquad u \in \mathcal{D}'(\mathbb{R}^2),\ \varphi \in \mathcal{D}(\mathbb{R}^2).$$

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